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Taxing Income Splitting Within Business-Owning Families*

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Abstract

Business owners can reduce their family tax liability by paying dividends to lower-income spouses. Governments have long used tax rules to restrict this practice, but little is known about how families respond. Using linked personal, corporate, and payroll tax records, I study Canada's 2018 expansion of the Tax on Split Income rules. Among spouses reporting positive dividends, dividend income falls by 53% in the treated group relative to controls. Higher owner dividends offset 57% of the dollar decline in spouse dividends, and owner labour income also rises. The reform therefore primarily reallocates reported income within families. I estimate a spouse-dividend elasticity of 1.30 with respect to the relative net-of-tax return to paying the spouse rather than the owner. With access to the same income-shifting channel, families across income groups show similar estimated elasticities. This finding suggests that differences in access to tax planning may help explain the larger taxable-income elasticities often estimated for higher-income taxpayers.

JEL classification: D13, G35, H24, H25, H26

Keywords: dividend taxation, income shifting, tax avoidance, tax-planning strategies, family businesses, intrahousehold income allocation

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1 Introduction

Business owners have strong incentives to allocate income to family members to minimize their tax liabilities.[1] Governments have therefore adopted a variety of policies to limit tax-motivated income splitting within families. Joint taxation, rules that attribute income to the person who generated it, and higher tax rates on income allocated to family members all reduce the tax advantage from within-family income splitting.[2] Despite the longstanding use of this class of policies and extensive concern about family income splitting, there is little empirical evidence on whether these policies reduce income splitting or how business-owning families adjust when they are introduced. Because these policies alter the tax consequences of allocating income within the family, understanding how families respond is necessary to assess both their equity and efficiency effects.

I address this gap by studying Canada's 2018 expansion of the Tax on Split Income (TOSI) rules. Canada provides a useful setting because linked longitudinal personal, corporate, and payroll tax records from the Canadian Employer-Employee Dynamics Database (CEEDD) allow me to follow both spouses and their family business over time. Canada's progressive personal income tax is levied at the individual rather than family level. Before the reform, business owners could reduce their family's tax liability by issuing dividends to lower-income family members.[3] The federal government announced the proposed expansion of the TOSI rules in July 2017, and the new rules took effect on January 1, 2018. Dividends received from a related private corporation became subject to the top marginal tax rate unless the recipient qualified for an exemption based on the age of the business owner, ownership of the corporation, participation in the business, or a reasonable return on contributions of labour, capital, or risk. These exemptions provide a basis for comparing families affected by the reform with those that are not.

My first strategy estimates a difference-in-differences model comparing families classified as affected by the TOSI reform with families classified as unaffected. I use spouses' common-share ownership and labour income to approximate the 10% ownership and 20-hour weekly participation exemptions, and restrict primary owners to those under age sixty-five. Treated spouses remain below the observable ownership and employment thresholds, while controls meet at least one of these observable thresholds. The main sample is a balanced panel of couples who fully own their

[1] Evidence from the United Kingdom shows that the introduction of individual taxation led couples to shift investment income toward the lower-taxed spouse (Stephens & Ward-Batts, 2004). In Canada, Wolfson and Legree (2015) estimate that income splitting through private corporations reduced tax revenues by approximately $500 million in 2011.

[2] Examples include joint taxation of spouses in the United States and Germany, which generally limits the personal-income-tax benefit of reallocating income between spouses but does not extend to other family members; the clubbing and attribution rules in India and South Africa, which assign certain income transferred to family members back to the original taxpayer; the personal-services-income rules in Australia and New Zealand, which attribute qualifying income to the person whose services generated it; and Sweden's 3:12 rules, which restrict preferential taxation of dividends from closely held corporations. The United Kingdom's settlements rules attribute some transferred income to the transferor but permit dividends paid through genuine ownership interests in family businesses.

[3] Canadian corporations can pay different dividends to different classes of shares, allowing payments to be directed toward particular shareholders.

linked corporation, with the spouse holding shares in no other business. Spouse dividends initially rise by approximately 10.5% relative to controls in the 2017 announcement year, consistent with families bringing payments forward before the new rules took effect. I find that, relative to controls, treated spouses report 52.9% less dividend income after implementation on the intensive margin, measured among spouses reporting positive dividends. On the extensive margin, their probability of reporting any dividends falls by 16.8 percentage points. Both margins respond, consistent with evidence that dividend taxes affect payout amounts and the decision to pay dividends (Alstadsæter & Fjærli, 2009; Bilicka et al., 2025; Chetty & Saez, 2005). Primary owners report more dividends and total labour income, while wages paid to spouses by the family corporation fall. Much of the reduction in spouse income is therefore accompanied by higher income reported by the owner, although total couple dividends also decline. Family taxable income falls by only 1.2%, and family tax liability rises by 3.3%, with both log estimates significant at the 10% level and the dollar estimates imprecise. These much smaller family-level changes show why the decline in spouse dividends alone overstates the contraction of the family's personal tax base. The dividend response is larger among professional corporations, but is also present among non-professional corporations and families with shares in multiple businesses. A separate analysis of a broader sample that does not condition on exemption status shows little systematic movement in spouses' ownership or employment toward the observable exemption thresholds. Families primarily adjusted income payments, with little evidence of systematic changes in spouses' ownership or employment status.

The treatment–control comparison estimates the effect of the reform, but families experienced different changes in the tax incentive to allocate dividends to the spouse. My second strategy estimates how dividends respond to this relative tax incentive. I define the incentive as the log-difference between the spouse's and owner's net-of-dividend-tax rates. Because income choices also affect marginal tax rates, I instrument this incentive with the same measure calculated using fixed 2016 incomes and each year's tax rules. The estimated spouse-dividend elasticity is 1.30, so a 1% increase in the spouse-to-owner net-of-tax-rate ratio increases spouse dividends by approximately 1.30%. The probability of reporting dividends and the spouse's share of couple dividends also rise with the incentive, while owner dividends fall. Thus, when the reform reduces the advantage of paying the spouse, families reduce those payments and increase dividends reported by the owner. These responses complement evidence that business owners adjust income allocation when tax incentives change (Alstadsæter & Jacob, 2016; Harju & Matikka, 2016a). The IV estimates for the full sample show no statistically significant response in family taxable income or tax liability. I also find no statistically detectable response in assets or retained earnings. Other settings exhibit profit retention following dividend or personal tax changes (Alstadsæter et al., 2014; Miller et al., 2024), while Yagan (2015) finds no investment or employee-compensation response to the US dividend tax cut.

The spouse-dividend elasticities are similar across family-income quartiles, at 1.22, 1.14, 1.69, and 1.28 from Q1 to Q4, with the lowest-quartile estimate less precise. Families at different income

levels therefore show similar proportional responses to a comparable change in the incentive to split income. This finding offers a useful perspective on the larger taxable-income elasticities often estimated for high-income taxpayers (Gruber & Saez, 2002). One explanation for that pattern is that higher-income taxpayers have more opportunities to adjust how their income is reported (Saez et al., 2012; Slemrod & Kopczuk, 2002). Here, families throughout the income distribution have access to the same dividend-allocation channel. Although my estimate concerns spouse dividends rather than total taxable income, their similar responsiveness supports the view that differences in access to tax planning help explain differences in measured tax responsiveness across the income distribution.

I also compare families according to the potential tax cost of maintaining their pre-reform dividend allocation. I rank treated families by the additional tax implied by their fixed 2016 incomes under the expanded rules. The treatment–control estimates show that spouse-dividend declines are concentrated in the third and fourth penalty quartiles, with the highest-penalty quartile showing the clearest combination of lower family taxable income and higher tax liability. Ranking families by income does not reveal the same pattern because the penalty also depends on how much income was allocated to the spouse. Comparing both rankings therefore distinguishes families' economic resources from their use of the restricted channel.

My research makes several contributions. To my knowledge, I provide the first causal evidence on how a rule targeting dividend-based income splitting in private businesses changes income reported by both the business owner and their spouse. Canadian studies document income splitting among business-owning couples (Schuetze, 2006; Wolfson & Legree, 2015), while Bauer et al. (2015) examine restrictions on dividends paid to minor children. Research on the taxation of couples shows changes in the allocation of assets and non-wage income (LaLumia, 2008; Stephens & Ward-Batts, 2004). Buettner et al. (2019) further show that couples' tax-planning choices depend on how those choices affect the distribution of net income between spouses. Creedy and Gemmell (2020) explain why joint decisions can alter individual taxable-income elasticities and revenue responses, and Alinaghi et al. (2023) find larger bunching-based elasticities among partnered individuals in New Zealand, particularly self-employed couples. I identify the response to a restriction on a specific dividend-allocation channel and follow both spouses and their corporation. This adds a family dimension to evidence on income shifting across corporate and personal tax bases, between income forms, through changes in business legal form, and across time (Gordon & Slemrod, 1998; Gorry et al., 2021; Harju & Matikka, 2016b; Le Maire & Schjerning, 2013; Pirttilä & Selin, 2011; Romanov, 2006; Sivadasan & Slemrod, 2008; Slemrod, 1995; Tazhitdinova, 2020). It also complements research on closing passive-loss shelters, offshore banking channels, and dividend-withholding loopholes (Casi et al., 2026; Johannesen & Zucman, 2014; Samwick, 1996). By measuring changes in the recipient, form, total amount, and taxation of family income, I show how restricting the targeted channel affects income that remains in the personal tax base. As Saez et al. (2012) emphasize, the revenue implications of a reported-income response depend on whether income is taxed elsewhere and at

what rate. The family-level evidence here captures an important part of this adjustment by following income reported by both spouses and changes in their combined tax liability.

Second, I contribute to the literature on responses to dividend taxation by distinguishing the amount of income allocated through a tax-planning channel from responsiveness to a comparable incentive. Dividend-tax studies document changes in payouts and the composition of owner income (Alstadsæter & Fjærli, 2009; Bilicka et al., 2025; Chetty & Saez, 2005; Harju & Matikka, 2016a). The similar instrumental-variables estimates of spouse-dividend elasticities across family-income quartiles provide a useful comparison with the larger taxable-income elasticities often estimated for higher-income taxpayers (Gruber & Saez, 2002). Taxable-income elasticities reflect the tax environment and the opportunities it provides for adjustment as well as preferences (Kopczuk, 2005; Saez et al., 2012; Slemrod & Kopczuk, 2002). Consistent with this view, Alstadsæter and Jacob (2017) show that participation in tax avoidance depends on incentives, access, and awareness, while Adam et al. (2021) document bunching among UK owner-managers that is generally absent among employees. Participation in tax planning and responsiveness once a channel is available are distinct. Larger potential savings can encourage greater use of a channel without implying a larger proportional response among families already able to use it. My sample compares families with access to the same dividend-allocation channel and finds similar proportional responses to changes in its tax advantage across income groups. The estimated elasticity concerns spouse dividends rather than total taxable income, but the result offers evidence that differences in opportunities to alter reported income can help explain differences in measured tax responsiveness.

Third, I contribute to research on how tax planning and its consequences vary across the income distribution. Prior work documents income shifting and retention among high-income business owners (Miller et al., 2024; Pirttilä & Selin, 2011; Ring et al., 2026), while Koivisto (2025) finds strong dividend-tax responses among lower-income private-firm owners. I extend this work by linking dividend responses to changes in family taxable income and tax liability, and by comparing families according to both income and financial exposure. These comparisons distinguish families' economic resources from their use of the tax-planning channel and show how the distribution of reform effects differs when families are ranked by income or by the potential cost of continuing to split income. Following income and tax payments together helps assess the distributional consequences of restricting tax planning without equating changes in reported income with changes in economic activity or welfare (Chetty, 2009).

The remainder of the paper is organized as follows. Section 2 describes the taxation of private-business income in Canada and the 2018 expansion of TOSI. Section 3 introduces the linked administrative data, treatment classification, summary statistics, and descriptive trends. Section 4 presents the DiD specification and main results. Section 5 presents the IV specification and main results. Section 6 compares responses by family income, tax penalty, business type, and multiple-business ownership. Section 7 concludes.

2 Institutional Setting

2.1 Taxation of Private-Business Income in Canada

In Canada, both the federal and provincial governments levy corporate and personal income taxes. For all but two provinces, provincial personal and corporate income taxes are administered by the federal government through the Canada Revenue Agency (CRA).[4] Although provinces independently determine their own tax rates and certain credits, taxable personal and corporate income is generally determined using a common federal tax base. Consequently, although statutory tax rates vary across provinces, the fundamental structure of Canada's corporate and personal income tax systems remains broadly consistent across the country.

Active business income earned by corporations is subject to corporate income tax before profits are distributed to shareholders. Qualifying Canadian-controlled private corporations (CCPCs) face a reduced corporate tax rate on active business income up to a prescribed business limit through the Small Business Deduction, while active business income above that limit is taxed at higher corporate rate.[5] Personal income taxes are levied on individuals rather than households and are subject to a progressive rate schedule. Employment income is fully included in taxable income and taxed at the individual's applicable marginal tax rate. Capital gains receive preferential treatment through a 50% inclusion rate, meaning that only one-half of a realized capital gain is included in taxable income. For the period I study, Dividend income receives separate tax treatment relative to wage/salary income because corporate profits have already been subject to corporate income tax before being distributed to shareholders. Unlike dividends, salaries and bonuses are deductible when calculating corporate taxable income and are therefore taxed only once in the hands of the recipient. Dividends, in contrast, are paid from after-tax corporate profits and face a gross-up and tax credit in the hands of the shareholder to neutralize the corporate tax credit already paid

This gross up and tax credit system for dividend income is designed to promote horizontal equity between individuals who earn business income through incorporated businesses, those who operate unincorporated businesses, and those who derive their income through employment.[6] As a result, owner-managers generally derive only modest tax advantages from changing the form in which business income is received. Nevertheless, incorporation continues to provide considerable flexibility over the timing, form, and recipient of business income. For example, owner-managers may defer personal taxation by retaining earnings within the corporation rather than immediately distributing profits. In addition to determining the timing and form of business income, owner-

[4]Quebec administers both its personal and corporate income tax systems, while Alberta administers its own corporate income tax. Despite these administrative differences, taxpayers in all provinces remain subject to the federal Income Tax Act.

[5]A Canadian-controlled private corporation (CCPC) is a private corporation resident in Canada that is not controlled, directly or indirectly, by one or more non-residents or public corporations, as defined under the Income Tax Act. Unless otherwise noted, all references to corporations throughout the paper refer to CCPCs.

[6]In Canada, income earned by unincorporated businesses is taxed as if it were labour income.

managers may also determine who ultimately receives that income. By structuring ownership of the corporation among family members, dividends may be distributed across those shareholders rather than solely to the primary business owner. This practice is commonly referred to as income splitting.

Canada's progressive personal income tax system and its use of the individual, rather than the household, as the tax unit make income splitting a potentially lucrative form of tax planning. Rather than receiving all dividend income themselves, owner-managers may instead distribute dividends to lower-income family members who hold ownership interests in the corporation. Because dividends are taxed in the hands of the recipient, shifting dividend income from an individual facing a higher marginal tax rate to a family member facing a lower marginal tax rate can reduce the total amount of personal income tax paid on a given amount of corporate income. Income splitting through private corporations had long been recognized as a concern within the Canadian tax system. In response to the use of corporate and trust structures to transfer investment income to minor children, the federal government introduced the TOSI Rules—commonly known as the "kiddie tax"—in the 1999 federal budget, with the rules applying beginning in the 2000 taxation year. The tax subjected certain dividend and other property income received by individuals aged 17 or younger to the top marginal personal income tax rate rather than the ordinary progressive rate schedule. The stated objective was to discourage arrangements that shifted income from higher-income parents to lower-income children and thereby undermined the progressivity of the personal income tax system (Department of Finance Canada, 1999).

The original TOSI rules restricted income splitting with minor children, but dividends paid to adult family members generally remained subject to the recipient's ordinary progressive tax schedule. Business owners could therefore continue to reduce their family's tax liability by paying dividends to lower-income adult relatives. In 2018, the federal government expanded the TOSI rules to restrict this remaining channel, subject to several exemptions.

2.2 The 2018 Tax on Split Income Reform

On July 18, 2017, the federal government announced a package of proposed tax reforms targeting tax planning through private corporations, identifying income sprinkling (referred to throughout this paper as income splitting) as one of three principal areas of concern (Department of Finance Canada, 2017c).[7] Following revisions announced in October 2017 (Department of Finance Canada, 2017a), the revised Tax on Split Income (TOSI) rules took effect on January 1, 2018. Under the expanded rules, dividend income received by an adult family member from a related private corporation would be subject to the highest marginal tax rate unless the recipient satisfied one of several statutory exemption criteria. For affected individuals, the reform therefore represented a substantial increase in the marginal tax rate applied to dividend income.

[7]The other two areas of concern were passive investment income earned within private corporations and the conversion of income into capital gains.

The expanded TOSI rules provide several statutory exemptions under which dividend income would continue to be taxed under the ordinary personal income tax system. The exemption criteria differed depending on the recipient's age. For individuals aged 25 and older, dividend income remained subject to the ordinary dividend tax rules if any of the following conditions were satisfied. (i) the recipient had been actively engaged in the business for an average of at least 20 hours per week during the portion of the year in which the business operated in the current taxation year or in any five previous taxation years; (ii) the recipient qualified for the excluded shares exemption; (iii) the owner of the business paying out the dividend was aged 65 or older; or (iv) the dividend represented a reasonable return on the recipient's labour or capital contributions to the business. To qualify for the excluded shares exemption, the recipient must generally directly own shares representing at least 10% of both the voting rights and value of the corporation. Canadian corporations generally issue common and preferred shares. Common shares typically carry voting rights, while preferred shares are typically non-voting.[8] The corporation must also satisfy several additional statutory conditions, including that it is not a professional corporation and that less than 90% of its business income is derived from the provision of services. With the exception of the reasonable return provision, the exemption criteria are based on objective statutory rules.

Individuals between the ages of 18 and 24 faced considerably more restrictive exemption criteria. In particular, the excluded shares exemption was unavailable, and the reasonable return provision was limited to prescribed factors such as contributed capital and labour. Consequently, fewer statutory exemptions were available to individuals aged 18 to 24 than to those aged 25 and older. The exemption criteria were intended to distinguish family members making genuine labour or capital contributions from those receiving dividends primarily for tax-planning purposes. The exemption criteria generate several observable margins that determine exposure to the reform. Some, such as age, are predetermined and cannot be manipulated. Others reflect pre-existing ownership structures or labour market participation within the corporation that could be altered. I exploit these statutory distinctions to identify the behavioural response to the expansion of TOSI.

3 Data

I utilize the Canadian Employer-Employee Dynamics Database (CEEDD) (Statistics Canada, 2022), which contains administrative tax records for the universe of Canadian individual tax filers and Canadian-controlled private corporations (CCPCs) owned by either individuals or other corporations.[9] The primary analysis focuses on the period from 2012 to 2020. A key feature of the

[8]The rights attached to common and preferred shares are determined by the corporation's articles and need not follow these conventional characteristics. Corporations may issue multiple classes of common or preferred shares with different voting, dividend, and liquidation rights. For example, preferred shares may carry voting rights, while a class of common shares may be non-voting. Accordingly, eligibility for the excluded shares exemption depends on the voting rights and fair market value associated with the shares held by an individual rather than whether those shares are formally classified as common or preferred.

[9]Corporations owned by trusts are excluded from the CEEDD.

CEEDD is a set of identifiers that enables linkages between the individual and corporate tax files, allowing individuals to be connected to the corporations in which they hold ownership interests, corporations to be connected through corporate ownership structures, and firms to be linked to their employees. These linkages allow me to establish longitudinal relationships between corporations, their individual owners, the family members of those owners, and their employees, while simultaneously observing outcomes at both the individual and corporate levels.

3.1 Information on Individuals and Firms

The individual-level data are drawn from the T1 Family File (T1FF), the T1 Personal Master File (T1PMF), and T4 Statements of Remuneration Paid. These records provide information on individuals' demographic characteristics and income sources, including age, gender, province of residence, employment income, dividend income, taxable income, and total income. They also contain a unique individual identifier and information on family relationships, including a family identifier, family type, an indicator for whether an individual is identified as a child within the family, and, when relevant, an identifier for the individual's spouse. The T4 records additionally identify the employer paying an individual's employment income. Together, these records allow me to follow individuals and their income over time and link them to family members and corporations.

I measure employment income using T4 earnings from all employers and, separately, earnings paid by the family corporation. Total T4 income captures changes beyond payments from the family business, including earnings associated with increased labour supply to outside employers. T4 income from the family corporation isolates changes in employment income paid through that business.

I measure dividend income using the cash amount reported on each individual's personal tax return, before the gross-up used to calculate taxable income. These records include dividends from all sources, whereas the corporate records report total dividends paid by the business without identifying how much each shareholder receives. The personal records therefore allow me to distinguish changes in spouse dividends from changes in owner dividends, including cases in which payments to one fall while payments to the other rise. Although these measures also include dividends from outside the family corporation, the treatment–control comparison accounts for changes in those dividends that are common to both groups. The identifying assumption requires that, absent the reform, these other dividend sources would not generate different trends between treated and control families.[10]

I use log income outcomes to express responses in proportional terms given the right-skewed income distribution and the concentration of income splitting among higher-income families (Bauer

[10]Personal income is reported by calendar year, while corporate dividends, assets, and retained earnings are reported by corporate fiscal year, which may end in a different month. Payments received within the same calendar year can therefore fall in different corporate reporting periods, making the timing of annual personal and corporate outcomes less directly comparable.

et al., 2015; Wolfson & Legree, 2015). The log-dividend specifications capture the intensive-margin response among individuals reporting positive dividends. To examine the extensive margin, I construct an indicator, $ClaimDiv$, equal to one if an individual reports any dividend income and zero otherwise. I construct this indicator separately for spouses and primary owners. All log-income specifications exclude observations with zero or negative values of the corresponding outcome.

At the couple level, I sum income across the primary owner and spouse to construct measures of dividend income, employment income, taxable income, and total income. I measure family tax liability by summing each spouse's total tax calculated in the personal tax records. This observed outcome is distinct from the simulated tax rates and liabilities used to construct the tax incentive and TOSI penalty. Together, these outcomes allow me to examine whether reductions in spousal dividends coincide with increases in other income received by the couple. In robustness analyses, I examine total dividends paid by the family corporation.

The corporate-level data are drawn primarily from the T2 Schedule 50 Shareholder Information (T2S50) and the National Accounts Longitudinal Microdata File (NALMF). The T2S50 provides information on the common and preferred shares held by each shareholder. Shareholders are identified as either individuals or corporations, with identifiers that allow their ownership interests to be linked to the corresponding individual or corporate records. The NALMF provides information on corporate financial outcomes and characteristics, including revenue, profits, assets, retained earnings, and four-digit North American Industry Classification System (NAICS) industry codes. Using the corporate financial records, I construct measures of asset growth and retained earnings. I measure asset growth as the year-over-year percentage change in the corporation's total assets. Because retained earnings are reported at the beginning and end of each fiscal year, I measure average retained earnings during the year as the mean of beginning- and end-of-year retained earnings. I also calculate the year-over-year percentage change in retained earnings.

I use the Canadian Tax and Credit Simulator (CTaCS) to calculate marginal tax rates using each individual's province of residence, taxable income less cash dividends, cash dividend income, and their spouse's taxable income.[11] I calculate the dividend marginal tax rate as the change in simulated family tax liability resulting from a $100 increase in an individual's cash dividend income, divided by 100.[12] For the tax-incentive IV, I calculate rates using actual incomes in each year and construct simulated rates by holding both spouses' incomes fixed at their observed 2016 values while applying each year's tax rules. Both calculations incorporate TOSI for treated spouses after

[11] Eligible and non-eligible dividends receive different gross-ups and dividend tax credits, with non-eligible dividends generally facing higher personal tax rates. This distinction reflects the corporate tax paid on the underlying income. Eligible dividends generally correspond to income taxed at the general corporate rate, while non-eligible dividends generally correspond to income taxed at a lower rate through provisions such as the small business deduction. This deduction generally applies to qualifying active business income up to $500,000. The majority of firms in my sample have profits below $500,000, so I simulate tax liabilities assuming all dividends are non-eligible. To the extent that individuals receive eligible dividends, this assumption may overstate their simulated personal tax liabilities.

[12] Appendix A.1 describes the CTaCS grid and marginal-rate calculation.

implementation. No dividends are reassigned between spouses. Section 5.1 defines the actual and simulated incentives. For descriptive comparisons, I also calculate the spouse's marginal rate after setting their dividend income to zero, while leaving the owner's income unchanged.[13]

3.2 Linking Owners, Families, and Firms

I combine family, ownership, and employer identifiers to link each couple to its family corporation. Family identifiers connect individuals to their spouse, while the T2S50 shareholder records identify the corporations in which each spouse holds common or preferred shares. Employer identifiers on T4 slips then allow me to determine whether either spouse works for the family corporation and to measure the employment income it pays them. Together, these linkages allow me to follow income reported by both spouses alongside the ownership, employment, and financial outcomes of their business.[14]

3.3 Measuring Exposure to the TOSI Reform

The exemptions introduced with the expansion of the TOSI rules depend in part on the spouse's ownership interest in and employment by the family corporation. I use these two dimensions to classify spouses according to their likely exposure to the TOSI reform. The ownership-based exemption generally requires the spouse to own at least 10% of the corporation's votes and value. The employment exemption generally requires the spouse to work at least 20 hours per week in the business. Because the administrative data report employment income rather than hours worked, I approximate the employment condition using an annual earnings threshold equal to 20 hours per week for 52 weeks at the applicable provincial minimum wage.

Within the age-restricted sample described below, I classify a spouse as *likely exposed* if they held less than 10% of the corporation's common shares in every observed year from 2012 through 2020 and never earned employment income from the corporation at or above the earnings threshold in any year from 2013 through 2020. I classify a spouse as *likely exempt* if they held at least 10% of the common shares in every observed year or earned employment income from the corporation at or above the threshold in every year from 2013 through 2020. All spouses in the analysis sample hold common or preferred shares and are therefore eligible to receive dividends directly from the corporation. Spouses whose ownership and employment histories satisfy neither definition are excluded from the treatment–control comparison. I refer to likely exposed spouses as the treated group and likely exempt spouses as the control group in the empirical analysis.

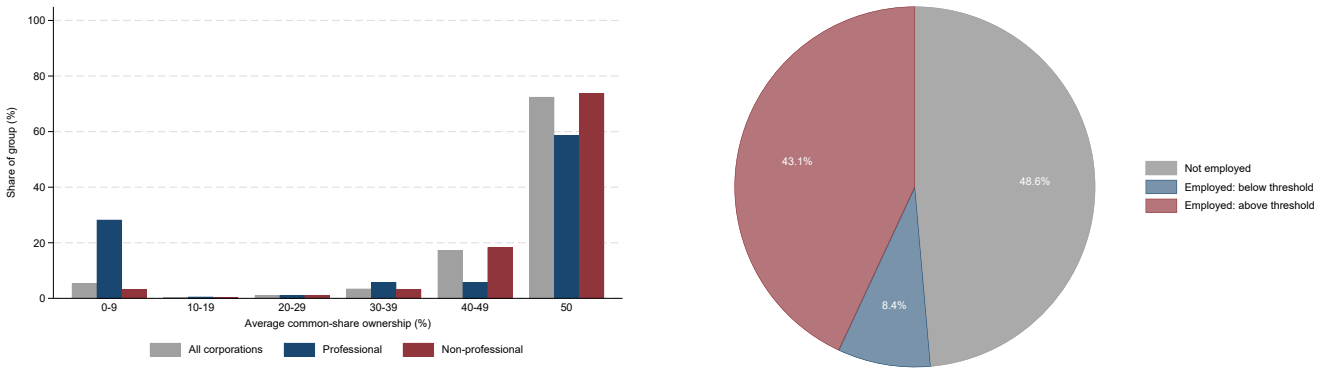
Figure 1 summarizes the pre-reform ownership and employment characteristics underlying this classification. Panel A shows that common-share ownership is concentrated at 50%, but low own-

[13]Personal dividend income includes payments from all sources. This counterfactual removes all dividends reported by the spouse, rather than only dividends from the family corporation.

[14]The family records also identify children living at the same address as their parents, but generally do not identify children living elsewhere or relationships between siblings. I therefore focus on income splitting between spouses.

ership is substantially more common among spouses associated with professional corporations. Approximately 28% of spouses associated with professional corporations hold less than 10% of the common shares, compared with approximately 3% of spouses associated with non-professional corporations. Panel B shows that approximately 49% of spouses are not employed by the family corporation, 8% are employed but earn less than the threshold, and 43% earn more than the threshold. The ownership and employment conditions therefore identify distinct sources of potential exposure to the reform.

Figure 1: Pre-Reform Ownership and Employment Characteristics



(a) Common-share ownership (b) Employment relative to the earnings threshold

Notes: Panel A reports the distribution of the spouse's average common-share ownership through 2016 for the full sample and separately for professional and non-professional corporations. Panel B classifies spouses using their average pre-reform employment income from the family corporation. The earnings threshold equals the amount earned by working 20 hours per week for 52 weeks at the applicable provincial minimum wage. "Not employed" denotes spouses with no employment income from the corporation. The figure is descriptive; the treatment classification uses the annual ownership and employment histories described in the text. Counts are subject to the rounding requirements applied to the administrative data.

The main classification requires spouses in the treated group to remain below both observable exemption margins throughout the sample period. This restriction excludes spouses who appear exposed before the reform but subsequently move above either the ownership or employment threshold. Appendix Table A3 compares this definition with an alternative that identifies the treated group using only pre-reform characteristics. Requiring treated spouses to remain below both thresholds reduces the potential treatment–control sample from approximately 64,460 to 63,790 families. The number of potentially treated families falls from approximately 3,700 to 3,030, while the control group is unchanged.

Appendix Figure A1 examines whether this reduction reflects systematic adjustment toward the observable exemption margins. The figure uses the broader sample of spouses eligible to receive dividends and defines exposure using only the age-based rules. It therefore does not condition on spouses' ownership or employment choices. The figure shows little evidence of systematic post-reform movement into the ownership or employment categories associated with an exemption.

These classifications measure likely exposure rather than establish each spouse's statutory tax treatment with certainty. Common-share ownership does not reveal whether a spouse owns at least

10% of both the votes and value of the corporation, and the availability of the ownership-based exemption also depends on characteristics of the business. Employment income similarly does not reveal hours worked. A spouse earning more than the provincial minimum wage could exceed the earnings threshold while working fewer than 20 hours per week. The classification also does not measure every potential exemption, including whether dividends represent a reasonable return. As a result, some affected spouses may be assigned to the control group and some exempt spouses may be assigned to the treated group. Such misclassification would generally make the two groups more similar and attenuate the estimated differences between them.

I assess the sensitivity of the results to this classification using alternative definitions based on common-share ownership alone and on common-share ownership among spouses with no recorded employment in the family corporation. Both alternatives yield declines in spouse dividend income and receipt alongside increases in primary-owner dividends, consistent with the main specification. Appendix A.5 describes these definitions and reports the estimates.

3.4 Sample Construction

My primary analysis examines income splitting between spouses. I restrict the sample to married or common-law individuals whose spouse can be identified in the administrative records. Within each couple, I designate the individual who owns the larger share of the corporation's common shares in 2016 as the primary owner and refer to the other individual as the spouse. If both individuals own equal shares, I assign these roles using a fixed ordering of individual identifiers. I hold the designations fixed throughout the analysis so that subsequent changes in ownership, employment, or income do not alter each individual's role within the couple.

Appendix Table A1 reports the cumulative sample restrictions. The linked data initially contain 1,114,185 married or common-law couples associated with at least one business. After requiring the variables needed to calculate tax rates to be observed and valid, the sample contains 1,078,135 couples. Among these couples, 397,585, or 36.9%, have observed share ownership for both spouses. I impose this restriction because an individual must own shares to receive dividends directly from the family corporation.[15] I next check the completeness of the ownership linkages by comparing the corporations I identify through the shareholder records with an administrative count of the corporations associated with each individual. I retain individuals for whom these counts agree. Appendix A.2 describes the ownership-linkage checks.

For the primary analysis, I further restrict the sample to spouses who hold shares in only one business. This reduces the sample from 350,510 to 264,060 couples. I require the corporation associated with the spouse to be owned directly by individuals and to have no ownership links to other corporations. The primary owner may hold shares in additional businesses; the restriction applies

[15] A family trust may hold shares in a private corporation and allocate dividend income to beneficiaries who do not directly own the shares. The CEEDD does not contain the information required to link family trusts to their beneficiaries, so I cannot identify income splitting conducted through these arrangements.

only to the spouse and the corporation in which the spouse holds shares. These restrictions identify a single family corporation from which the spouse may receive dividends and reduce ambiguity about the business from which changes in the spouse's income originate. I report estimates that include spouses who hold shares in multiple businesses as a robustness exercise.

I also require members of the family to collectively own 100% of the corporation's common shares. Because common shares typically carry voting rights, this restriction focuses the analysis on corporations whose voting control is likely to rest entirely with the family. To keep age-based eligibility fixed during the post-reform period, I exclude spouses aged 23–24 in 2018 and families whose primary owner was aged 63–64 in 2018. Spouses aged 23–24 would cross the age-25 threshold during the post-reform period, while the spouse of a primary owner who reaches age 65 may become eligible for the age-based spousal exemption.

Finally, I construct a balanced panel of families observed throughout the analysis period. This restriction reduces the sample from 220,100 to 94,295 couples and prevents changes in sample composition from driving changes in average outcomes over time. I report estimates using an unbalanced panel as a robustness exercise in Appendix A.6. After retaining families classified as either likely exposed to or likely exempt from the TOSI reform, the final sample contains 63,790 couples. The final sample therefore contains approximately 16.0% of the 397,585 couples in which both spouses directly own shares. Outcome-specific estimation samples may be smaller when an outcome is missing or nonpositive and therefore excluded from a log specification.

Appendix Table A2 compares the final analysis sample with the broader population of business-owning couples in 2016. The restrictions select families in which the spouse has a stronger observable connection to the business. In the analysis sample, 52.5% of spouses receive dividends and 39.7% are employed by the family corporation, compared with 34.8% and 19.1%, respectively, in the broader population. The analysis sample also contains larger shares of likely service businesses and professional corporations. The estimates should therefore be interpreted as applying to business-owning couples whose ownership and income allocation can be observed clearly, rather than to all families associated with private corporations.

Table 1 reports 2016 summary statistics separately for families classified as likely exposed to and likely exempt from the TOSI reform. Section 3.3 defines these classifications in detail. The exposed group contains 3,030 couples, while the exempt group contains 60,755. Differences in common-share ownership, preferred-share ownership, and employment by the family corporation largely reflect the criteria used to construct the groups. However, the groups also differ substantially in business composition. Professional corporations account for 47.3% of the exposed group but only 7.2% of the exempt group.[16] Similarly, 70.1% of exposed families are associated with likely service businesses, compared with 32.3% of exempt families. This concentration may partly reflect

[16]Provincial rules restrict voting shares to some licensed practitioners. For example, Ontario permits family members to hold non-voting shares in physician and dentist corporations, while reserving voting shares for members of those professions. In the full analysis sample, exposed spouses hold 0.08% of common shares on average, compared with 47.9% among exempt spouses (Table 1).

businesses organized around the primary owner's professional skills, in which substantial income can be generated with limited participation by the spouse. Paying dividends to a spouse with little common-share ownership or employment in the business places such families in the exposed group, whereas businesses jointly owned and operated by both spouses are more likely to enter the exempt group. These differences may also influence how firms respond to the reform. Businesses relying primarily on the owner's skills may have different capital requirements and investment opportunities from businesses producing goods, making their composition relevant when interpreting the corporate outcomes. Exposed families also report higher incomes. Average family taxable income is $218,068 in the exposed group and $137,804 in the exempt group. The concentration of professional and service corporations among exposed families is consistent with at least part of this income difference, although the summary statistics do not establish how much of the gap is attributable to business type. I therefore report estimates separately for professional and non-professional corporations as a robustness exercise.

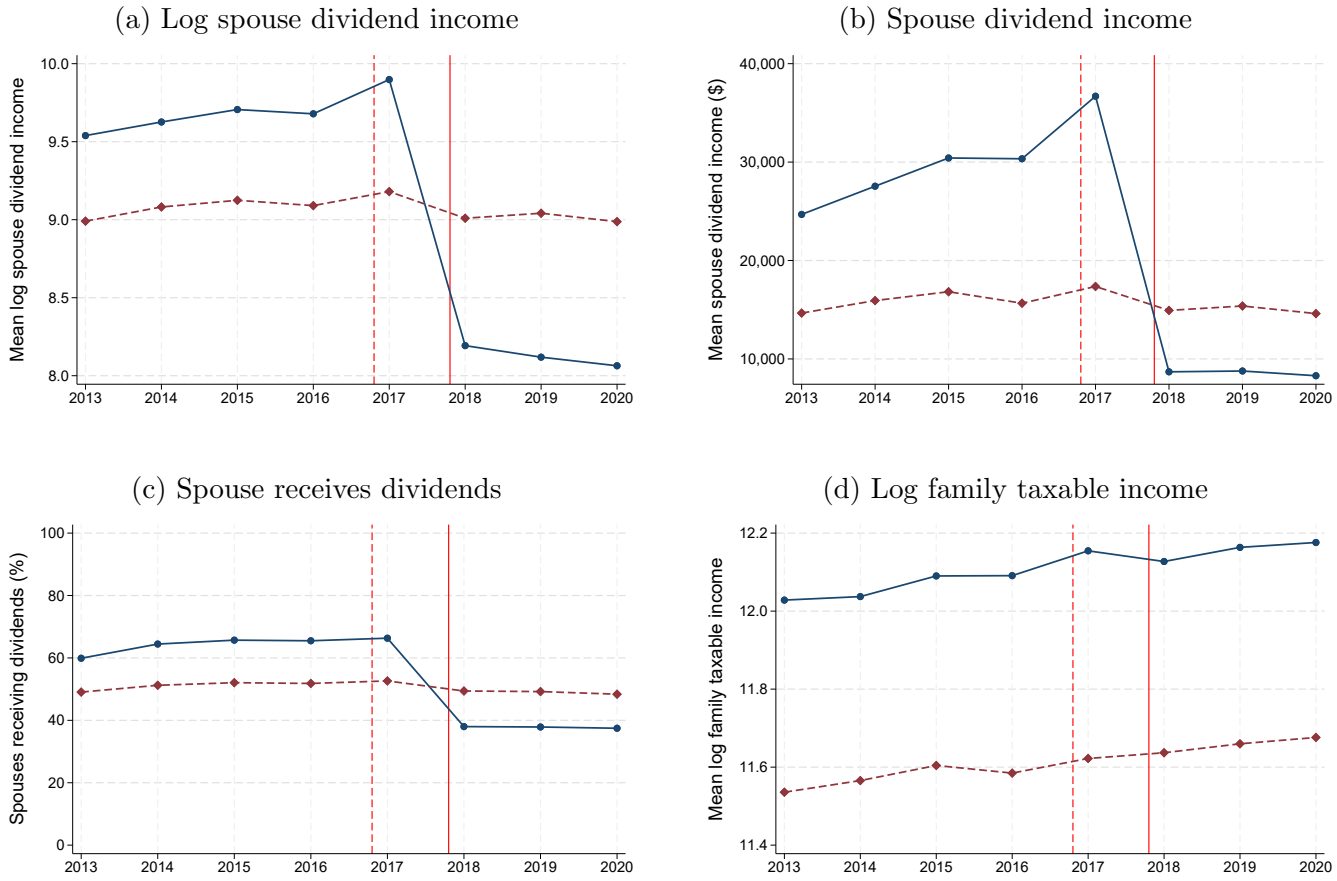
Table 1: Summary Statistics by Exposure to the TOSI Reform

	Likely exposed (N = 3,030)		Likely exempt (N = 60,755)	
Panel A: Demographics				
Spouse age (years)	47.1	(8.6)	49.2	(8.3)
Primary-owner age (years)	47.6	(8.1)	49.3	(7.7)
Panel B: Personal and family income				
Spouse receives dividends (%)	65.5	(47.5)	51.8	(50.0)
Spouse dividend income	30,339	(44,585)	15,661	(30,690)
Primary-owner dividend income	44,255	(58,430)	17,262	(38,815)
Spouse T4 income	55,060	(90,389)	43,304	(57,765)
Primary-owner T4 income	66,930	(103,477)	48,496	(84,448)
Family taxable income	218,068	(155,565)	137,804	(135,963)
Family tax liability	45,508	(62,963)	25,900	(52,532)
Panel C: Characteristics relevant to treatment classification				
Spouse common-share ownership (%)	0.08	(0.75)	47.90	(6.54)
Spouse preferred-share ownership (%)	67.18	(34.86)	6.85	(21.91)
Spouse employed by the family corporation (%)	2.87	(16.70)	41.53	(49.28)
Likely service business (%)	70.05	(45.81)	32.28	(46.76)
Professional corporation (%)	47.26	(49.93)	7.18	(25.82)
Panel D: Business characteristics and outcomes				
Business age (years)	7.6	(5.4)	11.6	(7.6)
Businesses in which the primary owner holds shares	1.24	(0.64)	1.11	(0.46)
Labour units	2.10	(8.52)	4.33	(9.23)
Gross profits	264,035	(371,359)	211,272	(407,863)
Total assets	487,274	(771,819)	556,787	(1,197,583)
Retained earnings	341,787	(577,038)	237,937	(607,660)

Notes: The table reports 2016 means, with standard deviations in parentheses, separately for families classified as likely exposed to and likely exempt from the TOSI reform. Monetary values are in Canadian dollars and are rounded to the nearest dollar. Sample sizes may differ for payroll and financial-statement variables because of missing data. Counts are rounded in accordance with the disclosure requirements applied to the administrative data; separately rounded group counts may therefore differ slightly from the rounded total sample count.

3.5 Descriptive Evidence

Figure 2: Spouse Dividends and Family Taxable Income Around the TOSI Reform



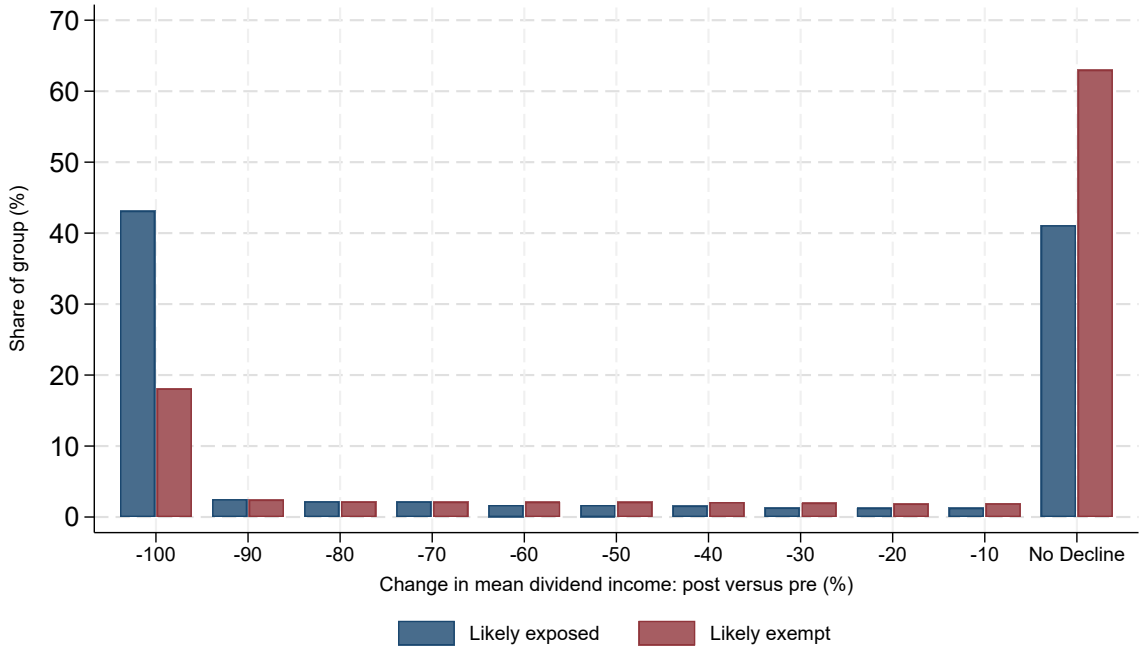
Notes: The figure reports annual means separately for families classified as likely exposed and likely exempt. The solid blue line with circles represents likely exposed families, while the dashed red line with diamonds represents likely exempt families. Panel A reports mean log spouse dividend income among spouses reporting positive dividends. Panel B reports mean spouse dividend income in dollars, including zero outcomes. Panel C reports the percentage of spouses receiving dividends. Panel D reports mean log family taxable income. The dashed vertical line marks the July 2017 announcement, and the solid vertical line marks the January 2018 implementation of the expanded TOSI rules.

Figure 2 shows annual income outcomes for treated and control families from 2013 through 2020. Treated spouses report more dividends and have higher family taxable income before the reform. Despite these differences in levels, spouse dividends levels, probability of receipt, and family taxable income follow broadly similar paths from 2013 through 2016. Spouse dividends rise among treated families in 2017 and fall sharply when the new rules take effect in 2018, while family taxable income shows a much smaller change. These descriptive patterns introduce the comparison examined in Section 4.2, where event-study estimates account for the included controls and fixed effects.

Figure 3 provides descriptive evidence on how spouse dividend income changed across the treatment and control groups defined in Section 3.3. For each spouse, I calculate the percentage change in average dividend income between the pre-reform period, 2012–2016, and the post-reform period, 2018–2020. The figure plots the distribution of these changes separately for treated and control

families. Treated spouses are substantially more concentrated in the category representing a complete elimination of spouse dividends. Control spouses are more likely to experience no decline. The remaining reductions are distributed across similar intermediate categories in both groups. This descriptive pattern is consistent with larger reductions in spouse dividends among families classified as treated.

Figure 3: Changes in Spouse Dividend Income by Exposure Classification



Notes: The figure shows the distribution of the percentage change in each spouse's average dividend income between the pre-reform period, 2012–2016, and the post-reform period, 2018–2020. Reductions are grouped into 10-percentage-point bins. The category labelled "No decline" includes spouses whose average dividend income did not decline. The sample is restricted to spouses with positive average dividend income during the pre-reform period. Bars report the share of treated or control spouses in each category. Group shares are calculated using vetted, rounded cell counts.

4 Effects of the TOSI Reform on Dividend Payments and Income Reallocation

In this section, I estimate the effect of the TOSI reform on dividends reported by spouses of business owners using a difference-in-differences design based on the treatment and control classifications described in Section 3.3. The reform did not directly change the tax treatment of dividends received by the primary owner. Families could therefore respond by redirecting dividends to the owner. I also explore whether the reform led to changes in labour income for either the spouse or the owner. Before the reform, additional labour income could raise the spouse's marginal dividend tax rate and reduce the family's tax savings from income splitting. By applying the top marginal

tax rate to covered spouse dividends, the reform weakened the incentive to keep the spouse's other taxable income low to preserve a lower dividend tax rate. Previous research shows that changes in family taxation can affect spousal labour supply (LaLumia, 2008). Dividend tax reforms can alter investment incentives (Yagan, 2015) and the incentive to retain income within the firm (Alstadsæter et al., 2014; Miller et al., 2024). I therefore also examine whether the TOSI reform influences corporate assets and retained earnings in Section 5.3.

4.1 Difference-in-Differences Specification

I begin by estimating an event-study specification to examine differences in outcome trends before the TOSI reform and the timing of subsequent responses. The identifying assumption is that, in the absence of the reform, outcomes for treated and control families would have evolved in parallel.[17] I estimate:

$$Y_{it} = \alpha + \sum_{\substack{y=2013 \\ y\neq 2016}}^{2020} \beta_y \left(\mathbb{1}[t = y] \times Treated_i\right) + \delta' X_{it} + \gamma_i + \lambda_{p(i,t),t} + \mu_{j(i,t),t} + \eta_{a^s(i,t)} + \kappa_{a^o(i,t)} + \varepsilon_{it}. \tag{1}$$

Here, i indexes couples and t indexes years. The dependent variable Y_{it} is an outcome for the spouse, primary owner, or couple, as defined in Section 3.1. $Treated_i$ equals one for families classified as treated and zero for families classified as control. The term γ_i denotes couple fixed effects. The terms $\lambda_{p(i,t),t}$ and $\mu_{j(i,t),t}$ denote province-by-year and four-digit NAICS-industry-by-year fixed effects, respectively.[18] The terms $\eta_{a^s(i,t)}$ and $\kappa_{a^o(i,t)}$ denote spouse-age and primary-owner-age fixed effects. The vector X_{it} contains the number of jobs, the numbers of main and subsidiary businesses owned by the couple, the total preferred-share ownership recorded for the family corporation, the number of individuals with recorded preferred-share holdings, and an indicator for whether the family has children. The omitted reference year is 2016, the final year before the reform was announced. Each β_y measures the difference between the change in outcomes for treated and control families from 2016 to year y. The pre-reform coefficients provide evidence on whether the two groups followed similar trends before the reform. I cluster standard errors at the family level.

[17] In 2018–2019, the federal government reduced the small-business corporate tax rate and made associated changes to dividend gross-ups and tax credits (Department of Finance Canada, 2017b). These changes applied to qualifying businesses and shareholders in both groups, rather than following the spouse-specific TOSI exemption criteria. Although they could affect the magnitude of the estimates, the decline in treated spouses' dividends alongside higher owner dividends is consistent with income reallocation. The spouse-dividend response also begins in 2018, before the passive-income measures took effect for taxation years beginning after 2018 (Department of Finance Canada, 2018).

[18] Province-by-year fixed effects absorb annual shocks shared by families in the same province, including common effects of provincial tax reforms. They do not absorb tax changes that affect income brackets differently. Industry-by-year fixed effects absorb annual shocks common to businesses in the same industry, such as changes in commodity prices or differences across industries in their exposure to the COVID-19 pandemic.

I then estimate a more parsimonious difference-in-differences specification:

$$Y_{it} = \alpha + \beta_1 \left(Treated_i \times Announce_t\right) + \beta_2 \left(Treated_i \times Post_t\right) + \delta' X_{it} + \gamma_i + \lambda_{p(i,t),t} + \mu_{j(i,t),t} + \eta_{a^s(i,t)} + \kappa_{a^o(i,t)} + u_{it}. \tag{2}$$

The indicator $Announce_t = \mathbb{1}[t = 2017]$ identifies the announcement year, while $Post_t = \mathbb{1}[t \geq 2018]$ identifies the implementation period from 2018 through 2020. The coefficient β_1 measures the relative change in outcomes for treated families during the announcement year, and β_2 measures their average relative change after implementation. The controls and fixed effects are the same as in Equation (1). The identifying assumption is that, in the absence of the TOSI reform, the difference in outcomes between treated and control families would have remained stable after 2016.

4.2 Results

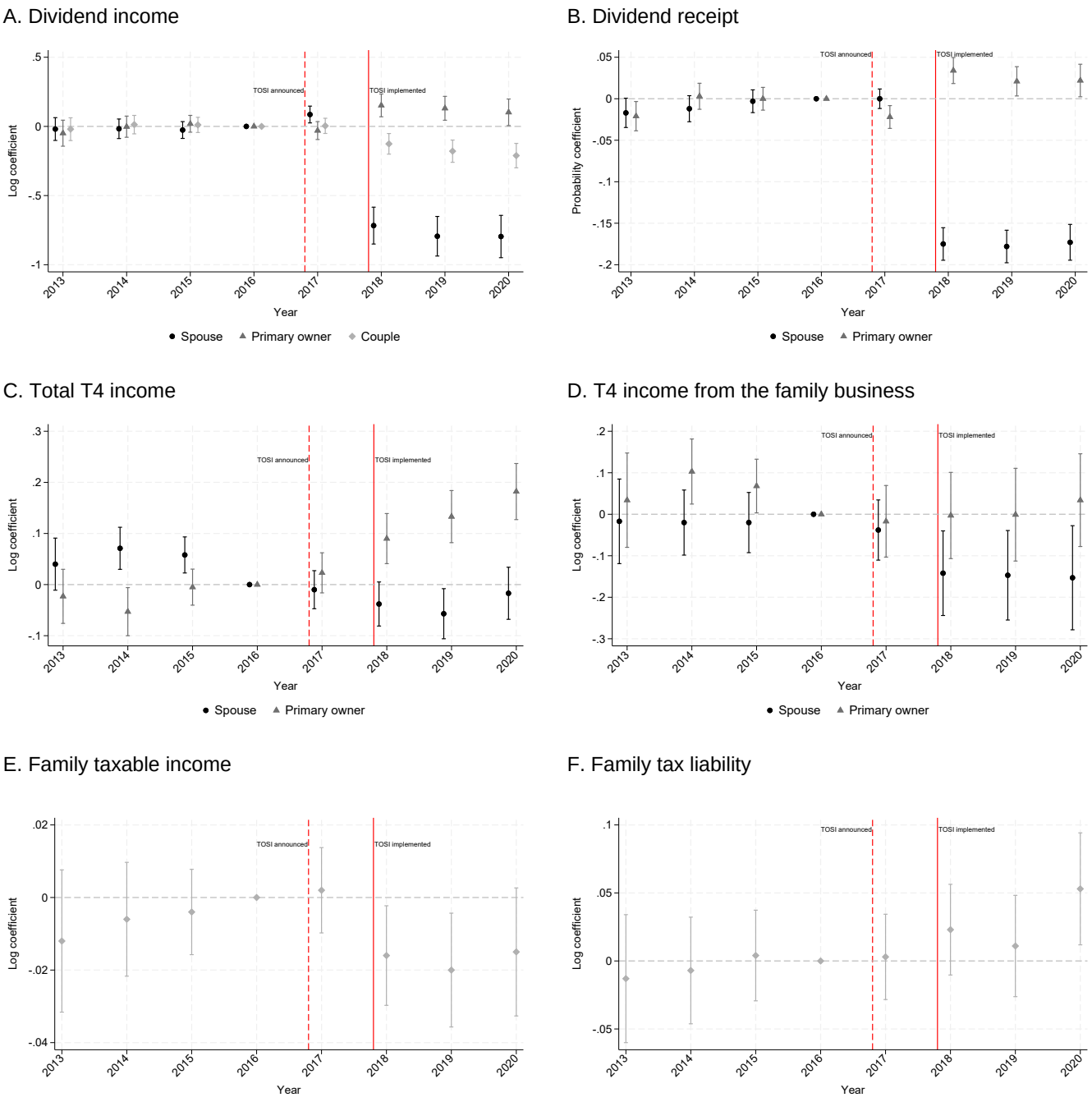
4.2.1 Dividend Income

Spouse dividends rise relative to the control group in 2017, consistent with families bringing payments forward before the TOSI reform took effect. They fall sharply in 2018 and remain lower through 2020 (Panels A and B of Figure 4). Primary-owner dividends rise after implementation, but couple dividends fall. The decline in the probability that treated spouses report any dividends shows that families respond by stopping some payments altogether, as well as by reducing the amounts paid to continuing recipients. The announcement-year increase is consistent with evidence that owners bring dividend payments forward before an anticipated tax increase (Alstadsæter & Fjærli, 2009).

After implementation, treated spouses report 52.9% less dividend income[19] and are 16.8 percentage points less likely to report any dividends than controls (Panel A of Table 2). Primary owners report 14.3% more dividends, and their probability of receipt rises by 2.9 percentage points. In dollars, the $12,391 decline in spouse dividends is partly offset by a $7,124 increase in owner dividends, leaving a $5,267 decline in couple dividends. The couple log estimate similarly implies a 15.9% decline. The PPML estimates, which include zero dividend outcomes, support the same pattern (Appendix B1). Higher owner dividends offset approximately 57% of the estimated dollar decline in spouse dividends. Changing the recipient therefore accounts for a substantial part of the dividend response, although couple dividends still fall. The owner remains available as an alternative dividend recipient, allowing families to reduce payments to the targeted spouse while preserving part of their dividend income. The decline in spouse dividends therefore overstates the reduction in dividends received by the family as a whole.

[19]The log-dividend regression excludes observations with zero dividend income; the dividend-receipt and dollar specifications include them.

Figure 4: Event-Study Estimates of Income Responses



Notes: Panels A–F report DiD event-study estimates from Equation (1). Panel A shows log dividends for spouses, primary owners, and couples; Panel B shows dividend receipt for spouses and owners. Panels C and D show log total and family-business T4 income for spouses and owners. Panels E and F show log family taxable income and tax liability. Circles denote spouses, triangles primary owners, and diamonds couple or family outcomes. The omitted year is 2016. Dashed and solid vertical lines mark the 2017 announcement and 2018 implementation. Bars show 95% confidence intervals based on family-clustered standard errors. Vertical scales differ across panels.

Table 2: Difference-in-Differences Estimates of Income Responses

Panel A. Dividend outcomes

	Log dividend income			Dividend receipt	
	Spouse	Primary owner	Couple	Spouse	Primary owner
Announcement, 2017	0.100***	-0.025	0.002	0.007	-0.018***
	(0.030)	(0.032)	(0.027)	(0.006)	(0.007)
Implementation, 2018–2020	-0.753***	0.134***	-0.173***	-0.168***	0.029***
	(0.066)	(0.037)	(0.035)	(0.009)	(0.007)
Observations	239,795	255,870	295,125	478,735	478,735

Panel B. Employment income

	Log total T4 income		Log T4 income from family business	
	Spouse	Primary owner	Spouse	Primary owner
Announcement, 2017	-0.051***	0.042**	-0.025	-0.066
	(0.018)	(0.021)	(0.039)	(0.047)
Implementation, 2018–2020	-0.079***	0.153***	-0.133***	-0.040
	(0.020)	(0.024)	(0.047)	(0.051)
Observations	377,555	374,695	215,465	177,420

Panel C. Family taxable income and tax liability

	Log family taxable income	Log family tax liability
Announcement, 2017	0.007	0.007
	(0.007)	(0.016)
Implementation, 2018–2020	-0.012*	0.032*
	(0.007)	(0.017)
Observations	478,260	456,090

Notes: The table reports the coefficients on Treated × Announce and Treated × Post from Equation (2). The announcement indicator equals one in 2017; the implementation indicator equals one during 2018–2020. Both coefficients are estimated jointly in the same regression. Dividend receipt is a zero-one indicator, so its coefficients are in probability units; all other outcomes are in logs. Each outcome has its own estimation sample. Family-clustered standard errors appear in parentheses. The regressions include the controls and fixed effects described in Section 4.1. Dollar and PPML estimates are reported in the appendix. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

4.2.2 Labour Income

Labour income provides another possible adjustment margin. Panel C of Figure 4 shows that spouses' total T4 income declines relative to controls before the reform and continues to decline afterward, without a clear break at implementation. Labour income from the family corporation follows a different pattern. Panel D shows a sharp decline beginning in 2018, with payments falling by 12.5% over the post-reform period (Panel B of Table 2). This is notable because the TOSI reform did not directly change the taxation of labour income. One possible interpretation

is that wage payments to spouses also served as an income-splitting channel, and business-owning families reduced their use of this channel after the reform to avoid attracting scrutiny from the Canada Revenue Agency, Canada's tax authority. The limited movement toward the ownership and employment thresholds associated with exemption is consistent with this interpretation (Appendix Figure A1), although it does not identify audit concerns as the explanation.

Primary owners' total labour income rises by 16.5% after implementation, alongside the increase in their dividend income. However, there is no statistically significant increase in labour income paid to owners by the family corporation in the log specification. Additional earnings outside the linked corporation could contribute to the total increase, but these regressions do not directly identify outside earnings or changes in hours worked. The results therefore show that the adjustment toward income reported by the owner extends beyond dividends, while leaving the source of the additional labour income unresolved.

Taken together, spouse dividend income falls by 52.9% ($12,391) and total labour income falls by 7.6% ($4,419), while owner dividend income rises by 14.3% ($7,124) and total labour income rises by 16.5% ($10,867). The dollar point estimates sum to a $16,810 decline for spouses and a $17,991 increase for owners, indicating offsetting changes of similar magnitude.[20] These patterns complement evidence that business owners change the wage–dividend composition of their income when relative tax rates change (Alstadsæter & Jacob, 2016; Harju & Matikka, 2016a). While those studies focus on the form of income received by an individual owner, my results highlight the additional role of the recipient within the family. Lower spouse income is accompanied by higher owner income in both dividends and labour earnings. I next examine family taxable income and tax liability to assess the combined consequences of these adjustments.

4.2.3 Family Taxable Income and Tax Liability

The fiscal consequences of the reform depend on both how much income the family continues to report and which spouse reports it. Redirecting income from a lower-taxed spouse to the owner can increase family tax liability even when total taxable income changes little. A reduction in reported family income can limit that increase. I therefore examine family taxable income and combined personal tax liability to assess whether the reform primarily changes the taxation of income that remains in the family tax base or also reduces the amount entering that base.

The event studies show little change in family taxable income in the 2017 announcement year and a modest decline after implementation (Panels E and F of Figure 4). Tax liability rises, although the annual increase is statistically clear only in 2020. Over 2018–2020, the estimates imply a 1.2% decline in family taxable income and a 3.3% increase in tax liability, both significant at the 10% level (Panel C of Table 2). The corresponding dollar estimates are imprecise (Appendix Table B1). Taxable income rises by $928 with a standard error of $1,875, and tax liability rises by $225 with a

[20] These totals sum separately estimated dollar coefficients; their uncertainty is not estimated jointly, and they do not trace the same payments between family members.

standard error of $755. The large reduction in spouse dividends is therefore accompanied by much smaller changes in the family's personal tax base and tax bill. Together with higher owner dividends and employment income, this pattern indicates that much of the adjustment changes the allocation and form of reported income within the family. This distinction matters for evaluating restrictions on tax planning. A decline in income reported through the targeted channel can substantially overstate the decline in income subject to personal taxation when other family members report more income. As Saez et al. (2012) emphasize, the revenue consequences of a reported-income response depend on whether that income is taxed elsewhere and at what rate. Following both spouses therefore provides information about the reform's fiscal consequences that the spouse-dividend response alone cannot provide.

4.2.4 Business Activity

The reduction in couple dividends raises the possibility that families retain more income inside the corporation rather than distribute it. I therefore apply the same difference-in-differences specification to corporate assets, retained earnings, and their annual growth rates. The estimates show no statistically significant change in retained earnings or either growth measure (Appendix Table B5). Total assets increase by 0.037 log points, approximately 3.8%, with the estimate significant at the 10% level. However, corporate assets and retained earnings were already rising relative to controls before 2016, and the event studies do not show a sustained increase beginning with implementation in 2018. The results therefore provide little consistent evidence of additional corporate accumulation corresponding to the decline in personal dividend income.

These findings complement the changes in income reported by family members. The reform leaves the owner available as an alternative dividend recipient, allowing families to reduce payments to spouses without necessarily retaining the same amount inside the corporation. Higher owner dividends and labour income, together with much smaller changes in family taxable income, make income reallocation the clearest observed response. The corporate estimates provide limited evidence of an accompanying balance-sheet adjustment, although they do not directly measure productive investment. One possible explanation is that families wish to maintain funds available for current household spending. Redirecting payments to the owner preserves access to that income, while retaining it inside the corporation offers tax deferral at the cost of postponing personal use of the funds. Several plausible explanations could account for the limited response in corporate assets and retained earnings. Families may find reallocating income to the owner less costly than retaining it inside the corporation and postponing personal access to the funds. The high proportion of professional and service-sector corporations may also help explain the limited asset response if these businesses have fewer opportunities for productive investment. I remain agnostic about the relative importance of these explanations.

The DiD results describe how the reform changed income reported by treated families. I next examine how reported income responds to the relative tax return from paying the spouse rather

than the owner.

5 Responses to Income-Splitting Incentives

The difference-in-differences results show how families adjust when the TOSI reform restricts income splitting. I next examine how sensitive these choices are to the tax advantage of paying a dividend to the spouse rather than the owner. The reform changed this advantage by different amounts across families, depending on the dividend tax rates each spouse would otherwise face. I use this variation to estimate the elasticity of spouse dividends with respect to the relative net-of-tax dividend rate and to examine how changes in this incentive affect income reported elsewhere in the family.

5.1 Tax Incentive and Instrumental-Variables Specification

I define the contemporaneous shifting incentive S_{it} as the log-difference between the spouse's and primary owner's net-of-marginal dividend tax rates,

$$S_{it} \equiv \log\left(1 - \tau_{it}^{S}\right) - \log\left(1 - \tau_{it}^{O}\right). \tag{3}$$

Here, τ_{it}^{S} and τ_{it}^{O} are calculated using each person's actual income in year t. From 2018 onward, the spouse rate incorporates TOSI for treated families, while control spouses remain subject to the ordinary dividend tax schedule. A larger value of S_{it} means that paying a covered dividend to the spouse leaves more income after personal tax relative to paying it to the owner. The TOSI rate describes the tax on an additional dividend from the related family business; it does not apply to unrelated portfolio dividends.[21]

Actual marginal tax rates depend on reported income, including the income choices being studied. I instrument S_{it} with an incentive calculated by holding all income inputs for both spouses fixed at their observed 2016 values and applying each year's tax rules,

$$Z_{it} \equiv \log\left(1 - \widetilde{\tau}_{it}^{S}\right) - \log\left(1 - \widetilde{\tau}_{it}^{O}\right). \tag{4}$$

The simulated rates $\widetilde{\tau}_{it}^{S}$ and $\widetilde{\tau}_{it}^{O}$ incorporate the year-t federal and provincial tax schedules, including TOSI for treated spouses after implementation. Both the spouse's and the owner's income inputs remain fixed. No dividends are reassigned between them. Changes in Z_{it} therefore reflect changes in tax rules applied to the pre-reform income allocation, rather than subsequent changes in income. This construction adapts the simulated-tax-rate approach of Gruber and Saez (2002) to the relative

[21] Personal tax records combine dividends from all sources. Applying the TOSI rate to the spouse's marginal family-business dividend therefore defines the relevant tax price for the restricted payment channel, rather than the tax rate on every dividend the spouse reports.

dividend-tax incentive faced by the couple. For descriptive comparisons, I also calculate the same fixed-income incentive without applying TOSI; this ordinary-tax counterfactual is separate from the actual-income incentive in Equation (3).

Appendix Figure C1 provides a separate descriptive comparison of pre-reform dividend amounts and sources across income and potential-incentive groups. The potential incentive used to form these groups removes spouse dividends without assigning them to the owner. It describes the tax-planning opportunity before those dividends are paid and is distinct from the actual-income incentive instrumented here.

The first and second stages are

$$S_{it} = \pi Z_{it} + \delta_S' X_{it} + \mathcal{F}_{it} + v_{it}, \tag{5}$$

$$Y_{it} = \beta \widehat{S}_{it} + \delta_Y' X_{it} + \mathcal{F}_{it} + e_{it}. \tag{6}$$

The term $\mathcal{F}_{it}$ collects the couple, province-by-year, four-digit-industry-by-year, spouse-age, and primary-owner-age fixed effects used in Equation (1); X_{it} contains the same controls. I retain treated and control families and cluster standard errors at the family level. When the outcome is log spouse dividend income, β is the elasticity with respect to the spouse-to-owner ratio of net-of-marginal dividend tax rates. When the outcome is dividend receipt, the coefficient instead measures a change in probability. Corresponding specifications for owner income and family outcomes show how these outcomes respond to the instrumented tax incentive. A causal interpretation requires that, conditional on the controls and fixed effects, changes in the simulated incentive affect the outcome through the contemporaneous incentive rather than other changes correlated with the instrument.

5.2 Results

Dividend payments respond strongly to the relative tax advantage of paying the spouse. The estimated spouse-dividend elasticity is 1.30 and is statistically significant at the 1% level, implying that a 1% increase in the spouse-to-owner net-of-tax-rate ratio raises spouse dividends by approximately 1.3% (Table 3). Owner dividends move in the opposite direction, with an elasticity of -0.342. These opposite signs are consistent with families directing dividends toward the recipient who retains more after tax. Because the TOSI reform reduces the spouse's relative net-of-tax return, it encourages families to reduce spouse dividends and increase payments to the owner. Owners receive larger dividend payments before the reform, so a smaller percentage response can represent a similar dollar change. The difference in elasticities therefore does not directly measure how much of the spouse's dividend reduction is offset by the owner. A 1% increase in the relative net-of-tax-rate ratio also raises the probability that the spouse reports any dividends by approximately 0.24 percentage points and the spouse's share of couple dividends by 0.23 percentage points. Adjustment thus occurs through both the amount of dividends reported and the decision to pay dividends to

the spouse.

The elasticity captures how spouse dividend income responds to the relative tax advantage of paying the spouse rather than the owner. This differs from a dividend-payout elasticity that measures how much income leaves the corporation. Families can respond strongly to the allocation incentive while continuing to receive much of the same income. These findings complement evidence that relative taxes influence owners' wage–dividend choices (Alstadsæter & Jacob, 2016; Harju & Matikka, 2016a) by showing that the identity of the recipient is another important margin of adjustment. This distinction helps explain why a sizeable spouse-dividend elasticity can accompany a small estimated response in family taxable income.

Table 3: Income Responses to the Relative Dividend-Tax Incentive

	Outcome					
	Log spouse dividends	Log owner dividends	Spouse dividend receipt	Spouse dividend share	Log family taxable income	Log family tax liability
S_{it}	1.299***	-0.342***	0.243***	22.773***	0.016	-0.076
	(0.193)	(0.118)	(0.026)	(2.336)	(0.026)	(0.068)
First stage	0.396***	0.508***	0.450***	0.491***	0.450***	0.462***
	(0.013)	(0.012)	(0.008)	(0.011)	(0.008)	(0.009)
First-stage F-statistic	975.70	1,927.00	2,932.00	2,072.00	2,932.00	2,950.00
Observations	239,735	255,785	478,540	295,030	478,070	455,970

Notes: The shifting incentive S_{it} is the contemporaneous log spouse-to-owner ratio of net-of-marginal dividend tax rates. The instrument applies each year's tax rules to both spouses' fixed 2016 incomes, including TOSI for treated spouses after implementation. No dividends are reassigned between spouses. All specifications include the controls and fixed effects in Equation (6). Family-clustered standard errors appear in parentheses. Observations and first-stage F-statistics are as released. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$. Dividend receipt is a zero-one indicator; the spouse dividend share is expressed in percentage points. Each column uses its own estimation sample and first stage. Coefficients on S_{it} for log outcomes are elasticities with respect to the relative net-of-tax rate. Additional average outcomes appear in Appendix Table C1.

The estimated responses of family taxable income and tax liability are much smaller than the dividend responses. Their elasticities are 0.016 and -0.076, respectively, and neither is statistically significant. These estimates are consistent with substantial adjustment in who reports income alongside much less change in the family's total taxable income. The tax-liability response reflects both the direct effect of the TOSI rules on dividends that spouses continue to receive and changes in the allocation of reported income.

Owners' total labour income also rises as the incentive to pay spouse dividends falls. The estimated elasticity is -0.138 and is statistically significant at the 5% level, implying that a 1% reduction in the spouse-to-owner net-of-tax-rate ratio raises owners' total T4 income by approximately 0.14%. Spouses' total T4 income shows no statistically significant response, and neither spouse nor owner labour income paid by the family corporation changes significantly. Higher owner labour income therefore accompanies the dividend reallocation, but the estimates do not show a corresponding increase in wages paid to owners by the linked corporation. As in the difference-in-differences results, distinguishing total labour income from payments by the family business is

important for understanding the adjustment. Appendix Table C1 reports these additional outcomes.

5.3 Business Activity

The tax-incentive analysis also examines whether changes in the return to paying spouse dividends affect corporate accumulation. Retaining income provides an alternative to distributing it to either family member. Miller et al. (2024) show that business owners use profit retention to shift income across time, while Alstadsæter et al. (2014) document substantial corporate retention following the introduction of dividend taxation in Norway. These findings motivate examining whether the loss of the spouse-dividend tax advantage leads families to accumulate more funds inside their corporations.

I find little consistent evidence of this adjustment. The estimated elasticities of retained earnings and corporate assets are 0.032 and 0.059, respectively, and neither is statistically significant. Annual asset growth and retained-earnings growth also show no statistically significant response (Appendix Table C1). These results reinforce the distinction between changes in income allocation and corporate accumulation. The tax incentive strongly affects dividends reported by spouses and owners, but the accompanying balance-sheet responses are much smaller and statistically insignificant.

The distinction between payout responses and business responses also appears elsewhere in the dividend-tax literature. Yagan (2015) finds that the US dividend tax cut increased payouts without detectable effects on investment or employee compensation. Bilicka et al. (2025) find that a dividend tax increase reduced payouts without increasing equipment purchases, while shifting investment toward assets potentially usable by owner-managers. My measures do not distinguish these asset categories or directly measure productive investment. The evidence nevertheless shows substantial adjustment in the recipient of reported income without a consistent accompanying increase in corporate accumulation.

6 Heterogeneity and Extensions

The distribution of responses matters for understanding who uses income splitting and who pays additional tax when it is restricted. Higher-income taxpayers often exhibit larger taxable-income responses to tax changes (Gruber & Saez, 2002), and differences in access to tax-planning opportunities are one explanation for this pattern (Saez et al., 2012). My setting allows me to examine responses among families at different income levels that could all allocate dividends to a spouse before the reform. I compare the DiD and IV estimates across family-income quartiles to examine both how much families adjust after the reform and how strongly they respond to a comparable change in the tax incentive. Following spouse dividends, owner income, family taxable income, and tax liability together also shows whether larger dividend responses translate into larger tax consequences. These distinctions are important for assessing how restrictions on income splitting

affect the progressivity of personal taxation.

Family income does not fully capture the potential cost of continuing to split income. That cost also depends on how much dividend income the spouse previously received and the increase in tax applied to it. A higher-income family that allocated little income to the spouse could face a smaller penalty than a moderate-income family that made greater use of this channel. I therefore also rank treated families by the additional tax they would owe under the TOSI rules if they maintained their observed 2016 income allocation. Comparing both designs across these groups examines whether the largest adjustments and tax consequences occur among the highest-income families or among those facing the greatest financial exposure. Finally, I examine whether the income-allocation results extend to professional and non-professional corporations and to spouses holding shares in multiple businesses.

6.1 Heterogeneity by Family Income

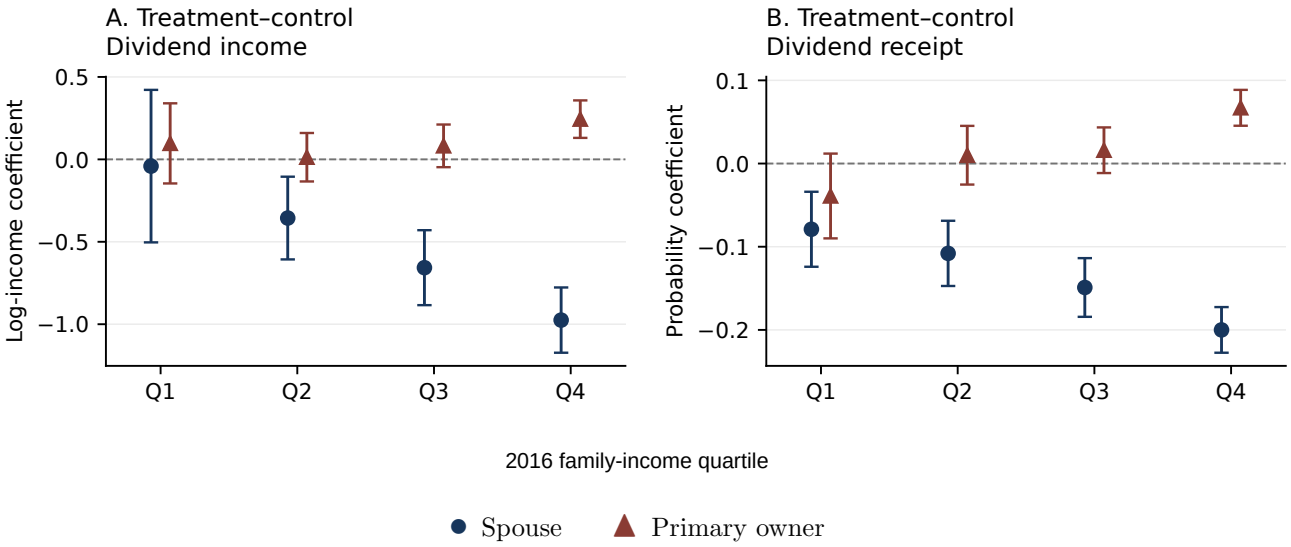
I divide families into quartiles of total family income in 2016, before the reform was announced, and hold these assignments fixed throughout the analysis. I first present the DiD results within each quartile, then use the IV estimates to examine how responsiveness to the relative tax incentive varies across the same income groups.

6.1.1 Difference-in-Differences Results

The decline in spouse dividends becomes larger at higher levels of 2016 family income (Figure 5). Spouse dividends fall by $2,015 in Q1 and by approximately 62% ($20,099) in Q4. The log estimate is imprecise in Q1 and becomes increasingly negative from Q2 through Q4. The probability of reporting any dividends also declines more at higher incomes, falling by 7.9 percentage points in Q1 and 20.0 percentage points in Q4, with both estimates significant at the 1% level. These results show adjustment throughout the income distribution, with larger dollar reductions per family at higher incomes. Only Q4 owners show a statistically significant increase in dividend income. Their dividends rise by $13,112, offsetting approximately 65% of the estimated dollar decline in spouse dividends. Redirecting dividends to the owner therefore accounts for a substantial part of the response among these families. Panels A and B of Appendix Table B2 report the underlying log and dollar estimates for all four quartiles.

The log estimates show higher owner total T4 income across all four quartiles, while a clear decline in spouses' total T4 income is concentrated in Q4. In that quartile, spouses report $9,535 less and owners $11,058 more in total T4 income (Appendix Table B3). Labour income therefore provides another part of the family response across income groups, alongside the increase in owner dividends concentrated at the top. However, no quartile shows a statistically significant increase in owners' labour income paid by the family corporation in the log specification. The family-level outcomes show a different pattern from spouse dividends across the income distribution (Figure 6).

Figure 5: Dividend Responses by 2016 Family-Income Quartile

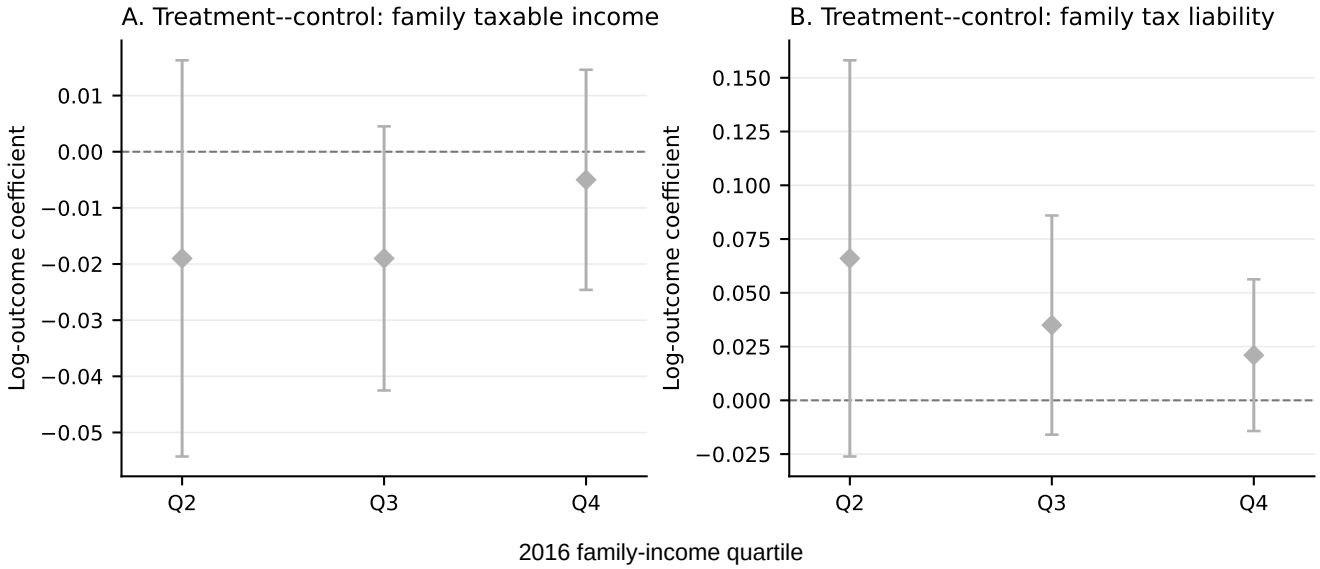


Notes: Panel A reports treatment–control estimates for log dividend income and Panel B reports estimates for the probability of reporting any dividends. Coefficients cover 2018–2020. Receipt coefficients are in probability units. Bars show 95% confidence intervals based on family-clustered standard errors. Appendix Table B2, Panels A and B, reports the corresponding log and dollar estimates.

The treatment–control comparison shows no statistically clear change in either log outcome in any quartile.[22] No quartile has a clear dollar increase in tax liability (Appendix Table B4). Taken together, the estimates show larger spouse-dividend reductions among higher-income families, accompanied by increases in income reported by owners, without clear evidence of correspondingly larger family tax increases.

[22]The dollar estimates for family taxable income are negative and significant at the 10% level for Q2 and Q3, despite imprecise log estimates. This may reflect large dollar reductions that are less apparent in proportional terms.

Figure 6: Family Taxable Income and Tax Liability Responses by 2016 Family-Income Quartile



Notes: Panels A and B report treatment–control estimates for 2018–2020. Both outcomes are in logs. Grey diamonds denote estimates and bars show 95% confidence intervals based on family-clustered standard errors. Q1 is omitted to make the other estimates easier to read. Appendix Table B4, Panels A and B, reports the log and dollar estimates for all four quartiles. Vertical scales differ across panels.

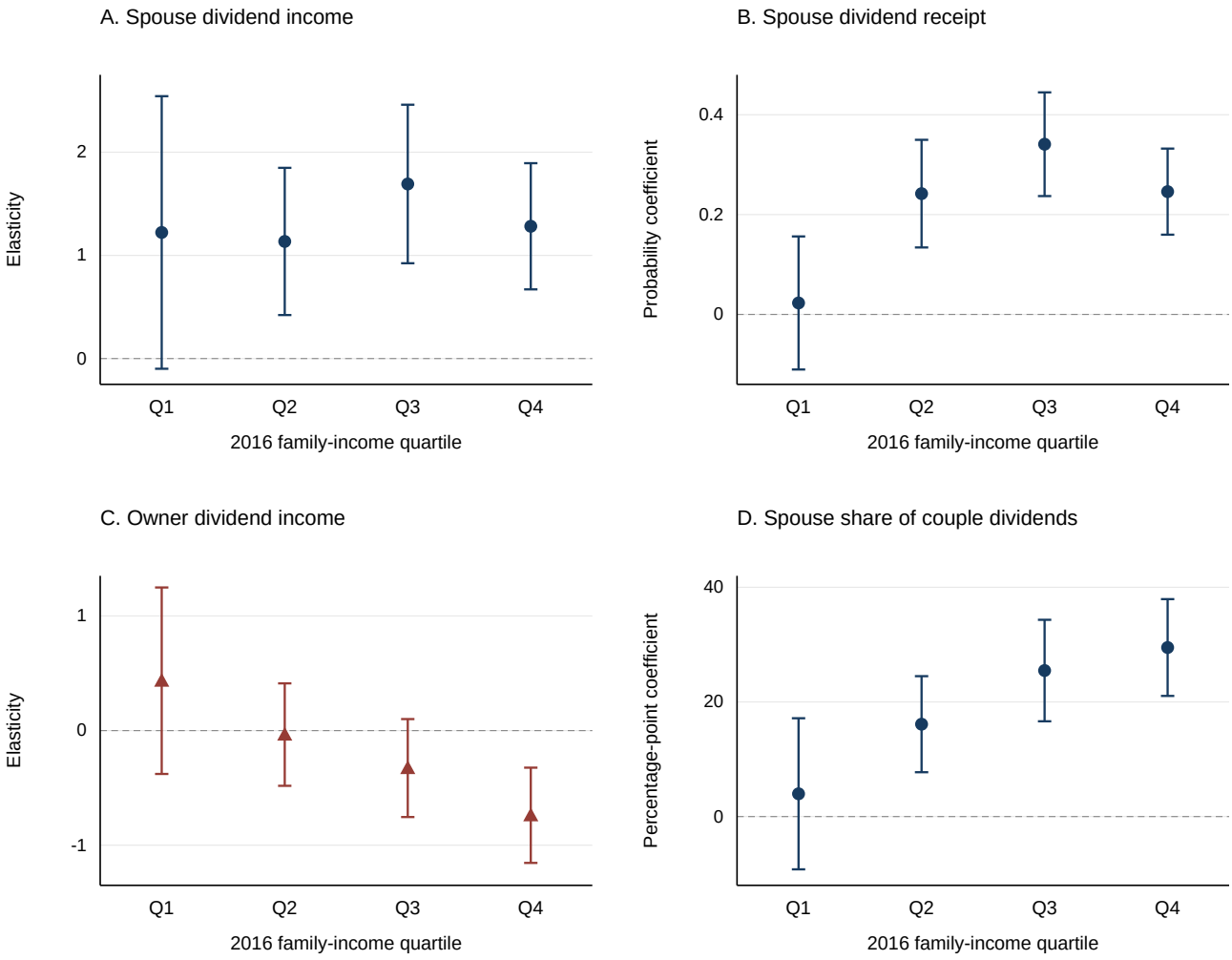
6.1.2 Instrumental-Variables Results

The estimated spouse-dividend elasticities are similar across the income distribution, at 1.22, 1.14, 1.69, and 1.28 from Q1 through Q4 (Figure 7). The lowest-quartile estimate is less precise and significant at the 10% level, while Q2–Q4 are significant at the 1% level. The estimates show no monotonic increase in responsiveness with family income. Appendix Table C2 reports the full estimates and their first stages.

These elasticities help interpret the larger dividend reductions among higher-income families in the DiD results. DiD measures the reform's effect within each income group, while IV measures the response to a comparable percentage change in the relative tax incentive. Actual incentive changes can differ across groups, allowing larger DiD declines to coexist with similar elasticities. Larger initial dividend payments also produce larger dollar changes for a given proportional response. The similar elasticities provide a useful comparison with the larger taxable-income elasticities often estimated for high-income taxpayers (Gruber & Saez, 2002). One explanation for those larger elasticities is greater access to opportunities for changing reported income (Kopczuk, 2005; Saez et al., 2012; Slemrod & Kopczuk, 2002). Families at each income level in my sample could allocate dividends to a spouse before the reform. Their similar estimated responsiveness supports the view that differences in tax-planning opportunities help explain income gradients in measured tax responsiveness, although the elasticity here concerns spouse dividends rather than total taxable income.

The probability of reporting any dividends also responds to the tax incentive. A 1% reduction in the spouse-to-owner net-of-tax-rate ratio lowers this probability by approximately 0.24, 0.34,

Figure 7: Tax-Incentive IV Estimates by Family-Income Quartile



Notes: Points show coefficients on the shifting incentive S_{it}, the actual log spouse-to-owner ratio of net-of-marginal dividend tax rates. The instrument applies contemporaneous tax rules to fixed 2016 incomes, including TOSI for treated spouses after implementation. Panels A and C show elasticities of spouse and owner dividend income. Panel B shows a probability coefficient for spouse dividend receipt. Panel D shows a coefficient for the spouse's share of couple dividends, measured in percentage points. All four income quartiles are retained. Bars show 95% confidence intervals based on family-clustered standard errors. Vertical scales differ across panels. Appendix Table C2 reports the matching first stages, observations, and F-statistics.

and 0.25 percentage points in Q2–Q4, respectively. The corresponding estimated reduction in the spouse's share of couple dividends rises from 0.04 percentage points in Q1 to 0.29 percentage points in Q4, with the Q1 estimate imprecise. The clearest owner-dividend response occurs in Q4, where the elasticity is −0.74, implying approximately 0.74% higher owner dividends for a 1% fall in the ratio. Owners' total labour income responds most clearly in Q2, rising by approximately 0.49% for the same incentive change (Appendix Table C4). Similar spouse-dividend elasticities therefore accompany different observed combinations of owner dividend and labour-income adjustments.

Family taxable income shows no statistically significant response in any quartile. For family tax liability, Q3 is the only group with a statistically significant response. A 1% reduction in the ratio

raises tax liability by approximately 0.15%, significant at the 10% level (Appendix Table C3). The IV estimates show no clear income gradient in the tax-liability response.

6.2 Heterogeneity by Tax Penalty

Family income and the potential cost of continuing to split income measure different aspects of exposure. I rank treated families by the additional personal tax they would owe under TOSI if they maintained their observed 2016 income allocation, averaging the simulated dollar penalty over 2018–2020. This calculation holds both spouses' income inputs fixed and changes only the tax treatment of the spouse's covered dividends; it does not move dividends to the owner. Appendix A.7 describes the calculation. I first report difference-in-differences estimates by penalty quartile, then use the same ranking to examine the tax-incentive IV. This separates where the policy had the largest effect from responsiveness to the relative tax incentive within groups facing different total stakes.

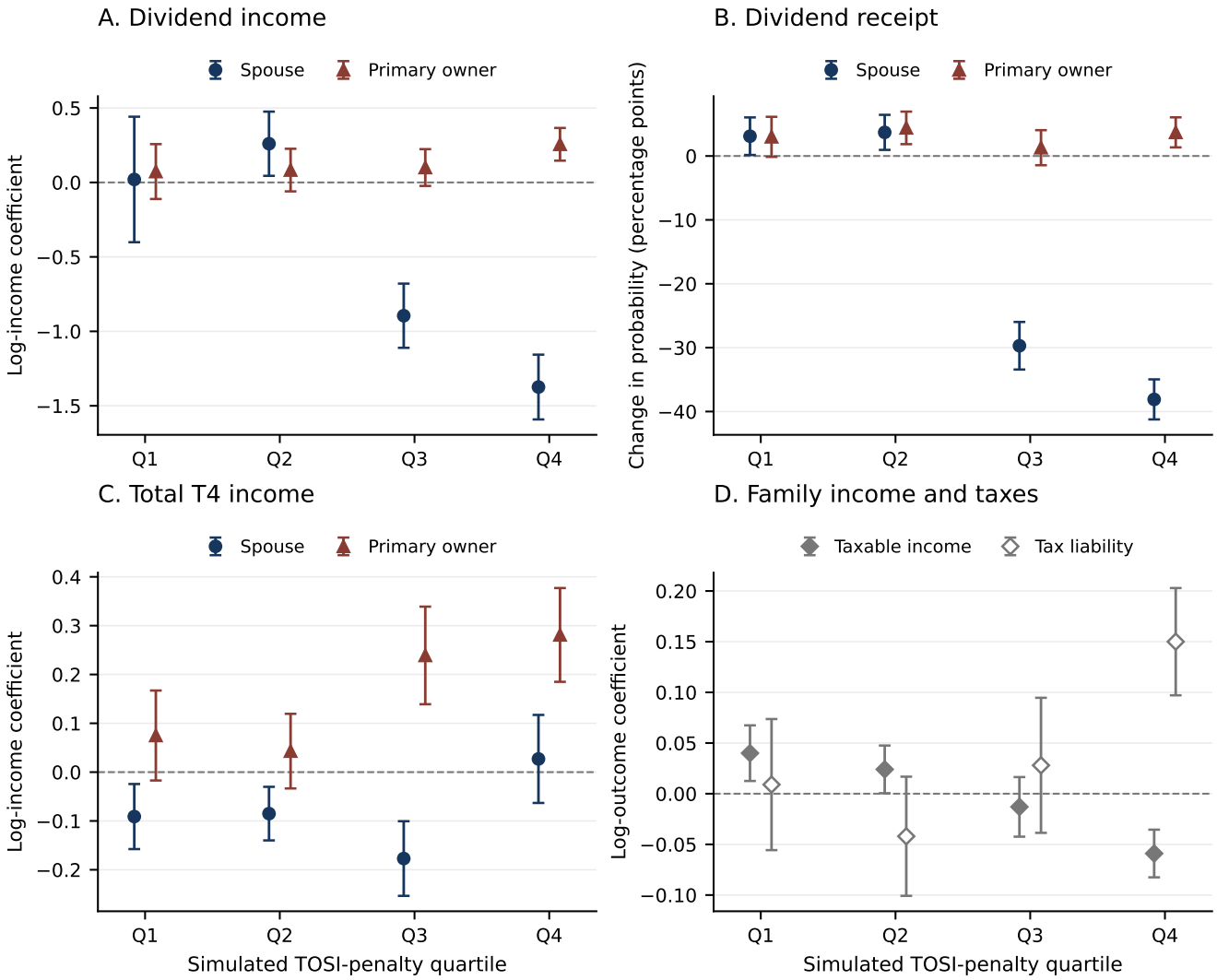
6.2.1 Difference-in-Differences Results

I compare treated families in each penalty quartile with the same control group. The resulting dividend responses differ sharply across quartiles (Figure 8). Log spouse dividend income shows no statistically significant change in Q1 and rises in Q2, but the estimates imply reductions of approximately 59% in Q3 and 75% in Q4. The probability that the spouse reports dividends falls by 29.7 percentage points in Q3 and 38.1 percentage points in Q4. The largest reductions occur where maintaining the pre-reform payment pattern would have been most costly.

The income and tax results show why ranking families by the amount at stake adds to the family-income analysis. Primary-owner dividends increase clearly only in Q4, while owners' total T4 income rises in Q3 and Q4. In Q4, family taxable income falls by approximately 5.7%, while tax liability rises by 16.2%. The dollar specification estimates a tax increase of approximately $4,840 (Appendix Table B7). Thus, income reported by the owner increases while the family's personal tax base contracts and its tax bill rises. This can occur if some income moves to a higher-taxed owner or if dividends still paid to the spouse face TOSI, even as less total income enters the personal tax base. The Q4 dollar estimate for taxable income is negative but imprecise, so the evidence for a decline in the tax base is stronger in proportional terms. Taxable income rises in Q1 and Q2 without a clear tax increase. In Q3, neither log family outcome changes clearly, although the dollar estimate shows a decline in tax liability of about $3,800. The clearest tax increase is therefore among families with the largest simulated penalties, a pattern that the ranking by family income alone did not reveal.

Labour-income results provide additional context. In the two highest penalty quartiles, owners' total T4 income rises and business-paid spouse T4 income falls, but there is no clear increase in wages paid to owners by the family corporation (Panel B of Appendix Table B6). The event studies

Figure 8: Treatment–Control Responses by Simulated TOSI-Penalty Quartile



Notes: Treated families are ranked into quartiles by their average simulated TOSI penalty for 2018–2020, calculated using fixed 2016 income. Each treated quartile is compared separately with the same control group. Points report treatment–control implementation-period estimates for 2018–2020. Panels A, C, and D use log outcomes. Panel B reports changes in the probability of receiving dividends, expressed in percentage points. Blue circles denote spouses and red triangles denote primary owners. In Panel D, filled grey diamonds denote family taxable income and hollow grey diamonds denote family tax liability. T4 income includes labour income from all employers. Bars show 95% confidence intervals based on family-clustered standard errors. Outcome samples and vertical scales differ across panels. Dollar estimates appear in Appendix Table B7; detailed log and receipt estimates appear in Appendix Table B6.

in Appendix Figures B1–B3 make the timing of the main results clearer. Spouse dividends in Q3 and Q4 had been moving toward the control group before the reform but fall sharply starting in 2018. Q4 family taxable income similarly reverses an earlier relative increase. These prior paths do not explain the post-reform declines. The Q4 tax-liability path had already been rising, so the increase in taxes has a less distinct break at implementation. In Q1 and Q2, the positive spouse-dividend estimates are already visible in 2017 and change little in 2018; I therefore do not interpret them as an increase caused by implementation.

6.2.2 Instrumental-Variables Results

The tax-incentive IV examines whether families facing larger potential penalties are also more responsive to a comparable change in the spouse-to-owner net-of-tax-rate ratio. The estimated spouse-dividend elasticities are positive in every quartile, at 1.07, 0.63, 0.94, and 0.47 from Q1 through Q4 (Appendix Table C6). They show no consistent increase with penalty rank. Q4, which has the largest DiD dividend reduction, has the smallest estimated elasticity. One possible explanation is that dollar responses to a given incentive change do not increase in proportion to the dividends initially paid. Families with larger dividend allocations must make larger dollar adjustments to generate the same percentage change in spouse dividends. Because the penalty ranking reflects both the tax change and the amount previously paid to the spouse, differences in the scale of these payments may contribute to the pattern in estimated elasticities. Owner-dividend responses are not statistically significant in any penalty quartile.

The family-level IV results are also mostly imprecise, with an exception in Q1. A 1% reduction in the net-of-tax-rate ratio implies approximately 0.13% lower family taxable income and 0.26% lower tax liability in this group, significant at the 5% and 10% levels, respectively. Neither family outcome responds significantly in Q2–Q4. These estimates show no clear pattern of larger tax increases as penalty rank rises (Appendix Table C6).

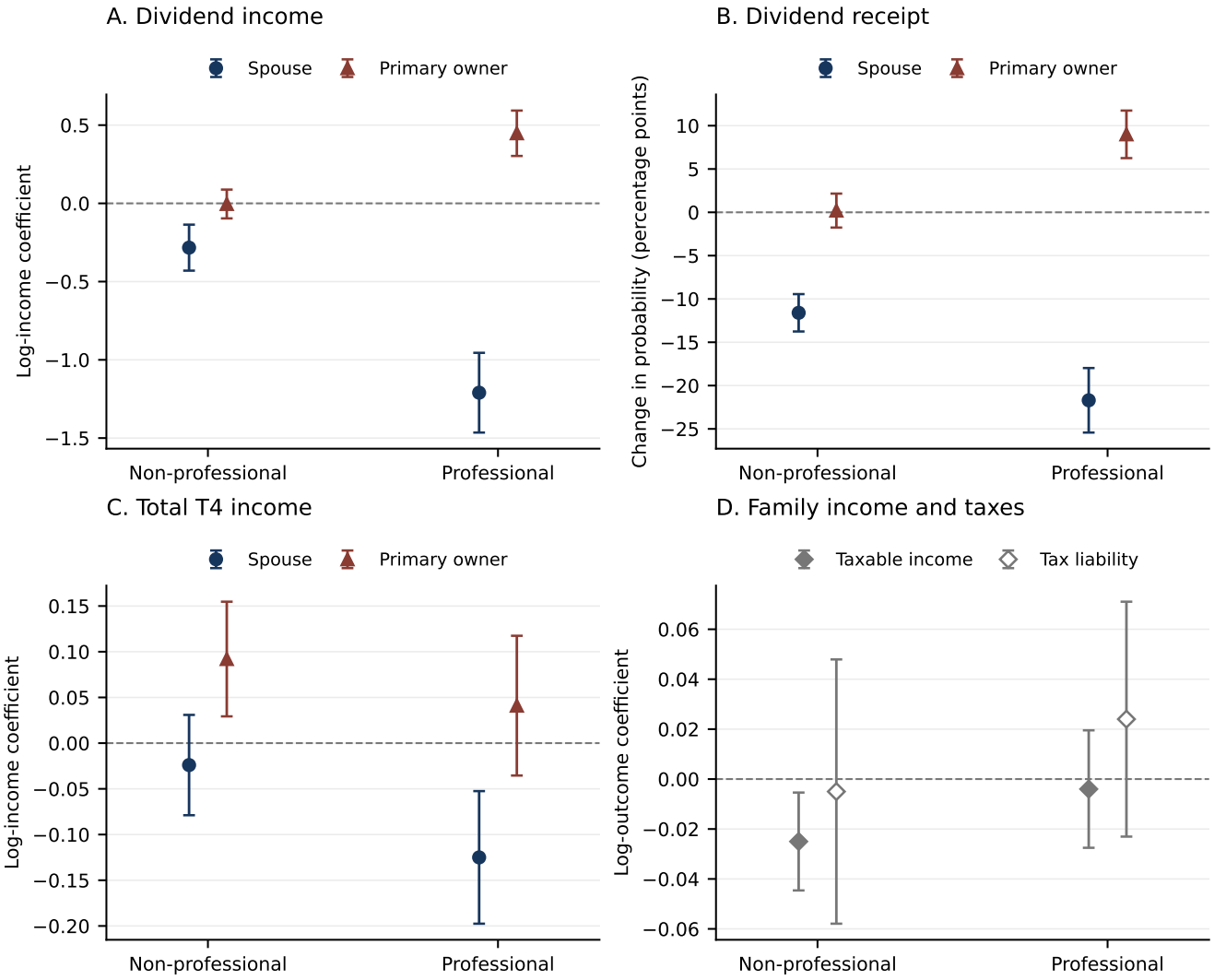
The largest DiD dividend reductions therefore occur among families facing the greatest financial stakes, without correspondingly larger estimated elasticities. The size of a policy response reflects the tax change and prior use of the channel as well as responsiveness to its incentive. Family income describes families' economic resources, while the penalty reflects the potential tax consequences of their existing dividend allocation. Comparing both rankings shows why evaluating restrictions on income splitting requires considering how much families used the channel as well as their income level.

6.3 Professional and Non-Professional Corporations

Professional corporations account for a substantially larger share of treated families than of controls. Their ownership rules also differ, and the 10% excluded-shares exemption is unavailable to professional corporations. I therefore examine whether their concentration explains the main results. Spouse dividends fall by 70.2% ($24,875) among professional families and 24.6% ($4,996) among non-professional families (Figure 9). The probability that a spouse reports any dividends falls by 21.7 and 11.6 percentage points, respectively. The larger professional response helps explain the magnitude of the main dividend decline, while the non-professional estimates show that the response extends beyond this group. Among professional families, owner dividends rise by $20,384, offsetting approximately 82% of the estimated dollar decline in spouse dividends. The log estimates also show different accompanying income changes. Owner dividends rise among professionals, whereas owners' total T4 income rises among non-professionals. Spouses' total T4 income falls among professionals.

The large professional dividend response occurs without a statistically significant decline in family taxable income, while non-professional families show a decline of approximately 2.5%. Neither group has a statistically significant DiD change in family tax liability. Appendix Table D1 reports the log, dollar, and dividend-receipt estimates.

Figure 9: Treatment–Control Responses by Corporation Type



Notes: Coefficients compare implementation-period changes between treated and control families separately within non-professional and professional corporations. Panels A, C, and D use log outcomes; Panel B shows changes in dividend receipt in percentage points. Blue circles denote spouses and red triangles denote primary owners. In Panel D, filled grey diamonds denote family taxable income and hollow grey diamonds denote family tax liability. T4 income includes all employers. Bars show 95% confidence intervals based on family-clustered standard errors. Outcome samples and vertical scales differ. Appendix Table D1 reports numerical estimates in logs and dollars and for dividend receipt.

The tax-incentive IV also shows a dividend response within both business types. The spouse-dividend elasticities are 1.82 for professional families and 0.83 for non-professional families, both statistically significant at the 1% level (Appendix Table C7). The professional owner-dividend elasticity is −1.26, implying approximately 1.26% higher owner dividends for a 1% reduction in

the spouse-to-owner net-of-tax-rate ratio. The owner-dividend response among non-professionals is not statistically significant. The probability that the spouse reports any dividends also responds significantly in both groups. Neither business type has a statistically significant IV response in family taxable income or tax liability. These comparisons describe reform responses within each business type, rather than the causal effect of professional status itself. Both designs show that income-allocation responses extend beyond professional corporations, with dividend reallocation toward the owner clearest among professionals.

6.4 Multiple-Business Ownership

Personal tax returns combine dividends from all sources. Restricting spouses to shares in one business makes these payments easier to relate to the family corporation. I relax this restriction to examine whether the response extends to families with more complex ownership structures. In the DiD comparison, spouse dividends among multiple-business families fall by 31.8% and the probability of reporting any dividends falls by 7.6 percentage points, compared with 52.9% and 16.8 percentage points in the main sample (Appendix Table D2). Owner dividend receipt rises in both samples, but owner dividend income rises significantly only in the main sample. The log estimates for multiple-business families also show higher owner total T4 income and lower labour income paid to spouses by the linked corporation. Family taxable income and tax liability show no statistically significant change.

For spouses holding shares in multiple businesses, a 1% reduction in the spouse-to-owner net-of-tax-rate ratio lowers the probability of reporting any dividends by approximately 0.18 percentage points and the spouse's share of couple dividends by 0.19 percentage points. These responses are significant at the 5% and 1% levels, respectively (Appendix Table C7). The estimated spouse-dividend elasticity is 0.22 and is statistically insignificant. Owner dividends, family taxable income, and family tax liability also show no statistically significant IV response. The dividend-receipt and allocation responses therefore extend to a broader ownership sample, while the elasticity of dividend amounts is less precisely estimated.

7 Conclusion

This paper examines how business-owning families responded to Canada's 2018 expansion of the Tax on Split Income rules, which restricted the tax advantage of paying dividends to family members. In the DiD comparison, spouse dividends fall sharply, while increases in owner dividends and labour income offset much of that decline. The log estimates show a modest decline in family taxable income and a modest increase in tax liability, while the corresponding dollar estimates are imprecise. The main observed response is therefore a change in who reports family income, showing why declines in payments to targeted spouses alone overstate the contraction of the family's personal

tax base.

Higher-income families have larger dollar reductions in spouse dividends, but the tax-incentive IV estimates similar elasticities across family-income quartiles. The elasticity for the full sample is 1.30, and the quartile estimates range from 1.14 to 1.69. Families at different income levels therefore respond similarly to a comparable change in the relative return to allocating dividends to the spouse. The taxable-income literature often finds larger elasticities among high-income taxpayers (Gruber & Saez, 2002), while responsiveness also depends on opportunities to adjust reported income (Kopczuk, 2005). My estimate concerns spouse dividends rather than total taxable income, but it examines a channel available to business-owning families at each income level. The similar elasticities suggest that differences in access to tax planning help explain the larger taxable-income responses often found at the top of the distribution. For the full sample, the IV estimates show no statistically significant response in family taxable income or tax liability.

Ranking families by the potential dollar penalty shows that the clearest increase in personal tax liability occurs among those facing the greatest financial exposure. The DiD estimates show the largest spouse-dividend declines in the third and fourth penalty quartiles, with family taxable income also falling in the fourth. Families with the highest incomes do not always face the largest penalties because the penalty depends on both the tax-rate change and the amount previously allocated to the spouse. The distribution of the reform's tax consequences therefore reflects families' use of the restricted channel alongside their income level.

Responses differ across business structures but extend beyond one type of family. In the DiD estimates, professional corporations account for much of the decline in spouse dividends, yet dividends also fall among non-professional families and among spouses with shares in multiple businesses. I find little consistent evidence of changes in corporate assets or retained earnings. The available corporate measures leave the destination of income missing from family tax returns unresolved, and the reported-income responses do not by themselves establish changes in real activity or welfare (Chetty, 2009). Evaluating this type of rule requires following income reported by both family members, the family's tax liability, and the business.

This paper focuses on a restricted sample that allows me to trace income flows between spouses, business owners, and a single family corporation. Future work could relax these restrictions to study a broader range of business structures and assess how income splitting affects the progressivity of Canada's personal income-tax system. The limited evidence of changes in business outcomes also raises the question of whether firms outside my sample respond differently to restrictions on income splitting. Finally, because income splitting gave some business owners an additional way to reduce their tax liability, it would be valuable to test whether owners who used this channel respond differently to broader tax changes than owners who did not, or could not, split income within their families.

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A Data and Treatment Classification

A.1 Tax Simulation Grid

I construct a lookup grid for marginal tax rates using the Canadian Tax and Credit Simulator (CTaCS). The grid covers 2010–2021 and 13 provincial and territorial tax jurisdictions, varying own income, non-eligible dividend income, and spouse income for a married tax filer. The own-income and dividend grids contain \$1, values from \$1,000 to \$50,000 in \$1,000 steps, values from \$55,000 to \$150,000 in \$5,000 steps, values from \$160,000 to \$200,000 in \$10,000 steps, and values from \$225,000 to \$350,000 in \$25,000 steps. The spouse-income grid contains \$1, \$10,000 to \$50,000 in \$10,000 steps, and \$75,000, \$100,000, \$150,000, \$200,000, \$300,000, and \$500,000. Spouse dividend income is set to zero in the grid inputs. CTaCS first calculates total family tax at each grid point. I then rerun it after separately adding \$100 to own earnings and to non-eligible dividends. Each change in total family tax divided by \$100 gives the corresponding earnings or dividend marginal tax rate. I match the resulting rates to the income allocations used in the analysis.

A.2 Sample Construction and Representativeness

I construct the broader linked sample using the individual and corporate shareholder identifiers in the T2S50. When a shareholder is itself a corporation, I trace ownership through the corporate identifiers to identify the ultimate individual owners. I retain corporations owned directly by individuals or indirectly through a single intermediate corporation. This allows conventional holding-company–operating-company structures but excludes ownership chains requiring more than one intermediate corporation to identify the individual owner.

I assess the completeness of these linkages using an administrative measure of the total number of corporations associated with each individual, either directly or indirectly. I compare this count with the number of corporations I independently identify through the ownership records and exclude individuals for whom the two counts differ. This check reduces the risk of treating an individual as associated with only one corporation when additional ownership interests are missing from the constructed linkages.

The main analysis imposes further restrictions on this broader sample. The spouse must hold shares in only one business, and that corporation must be owned directly by individuals and have no ownership links to other corporations. The primary owner may hold shares in additional businesses. Table A1 reports the cumulative sample restrictions, and Table A2 compares the final analysis sample with the broader population of business-owning couples.

Table A1: Construction of the Analysis Sample

	Cumulative sample restriction	Couples
0	Linked married or common-law couples with observed business ownership	1,114,185
1	Require treatment status to remain constant across observed years	1,112,625
2	Require the tax-rate variables used in the analysis to be observed and valid	1,078,135
3	Require observed share ownership	397,585
4	Exclude observations with incomplete corporate ownership links	369,765
5	Exclude families that cross a relevant age threshold	350,510
6	Require the spouse to hold shares in only one business	264,060
7	Require the family to own all common shares of that business	220,100
8	Require a balanced analysis panel	94,295
9	Retain families classified as likely exposed or likely exempt	63,790

Notes: Restrictions are cumulative and are applied in the order shown. One spouse identifies one couple and family in the analysis. Counts are rounded in accordance with the disclosure requirements applied to the administrative data. Additional outcome-specific restrictions may cause estimation samples to differ across regressions.

Table A2: Analysis Sample Compared with the Broader Population

	Broader population (up to N = 1,078,135)	Analysis sample (up to N = 63,790)
Panel A: Demographics		
Spouse age (years)	50.2 (13.2)	49.1 (8.4)
Primary-owner age (years)	51.6 (13.1)	49.2 (7.7)
Panel B: Personal and family income		
Spouse receives dividends (%)	34.8 (47.6)	52.5 (49.9)
Spouse dividend income	9,287 (41,124)	16,359 (31,643)
Primary-owner dividend income	20,674 (101,922)	18,545 (40,376)
Spouse T4 income	40,522 (80,356)	43,862 (59,772)
Primary-owner T4 income	52,018 (138,820)	49,372 (85,537)
Family taxable income	152,921 (278,060)	141,619 (138,017)
Family tax liability	31,703 (100,266)	26,832 (53,238)
Panel C: Ownership, employment, and business type		
Spouse common-share ownership (%)	37.96 (17.84)	45.62 (12.01)
Spouse preferred-share ownership (%)	12.80 (28.20)	9.72 (26.07)
Spouse employed by the family corporation (%)	19.1 (39.3)	39.7 (48.9)
Likely service business (%)	10.5 (30.7)	34.1 (47.4)
Professional corporation (%)	3.1 (17.3)	9.1 (28.7)
Panel D: Business characteristics and outcomes		
Business age (years)	10.8 (8.1)	11.1 (6.6)
Labour units	5.66 (22.74)	4.25 (9.21)
Gross profits	201,763 (1,342,287)	213,789 (406,348)
Total assets	875,169 (7,402,883)	553,480 (1,180,901)
Retained earnings	395,333 (4,217,809)	242,901 (606,632)

Notes: The table reports 2016 means, with standard deviations in parentheses. The broader population contains business-owning spouses in the linked administrative records before the restrictions used to construct the analysis sample. The analysis sample contains families classified as either likely exposed to or likely exempt from the TOSI reform. Monetary values are in Canadian dollars and are rounded to the nearest dollar. Sample sizes vary across variables because ownership, payroll, and financial-statement information is not available for every observation. The maximum sample reported here is based on observations with nonmissing information in the summary-statistics file and therefore differs from the unrestricted linked-data count reported in Table A1. Counts are subject to the rounding requirements applied to the administrative data.

A.3 Stability of the Exposure Classification

Table A3: Effect of Requiring a Fixed Exposure Classification

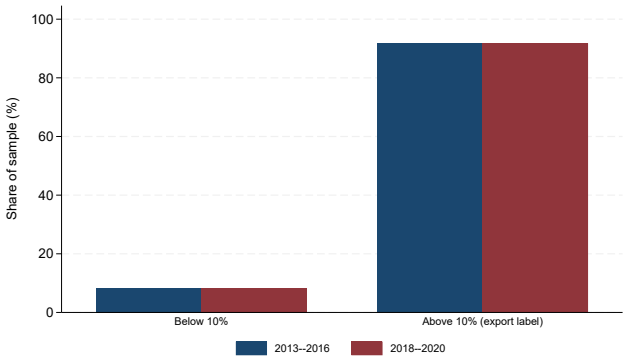
	Potential treated	Control	Combined
Before fixed-classification requirement	3,700	60,755	64,460
After fixed-classification requirement	3,030	60,755	63,790
Approximate number excluded	670	0	670
Share excluded	18.1%	0.0%	1.0%

Notes: The first row classifies spouses without requiring their ownership and employment classifications to remain fixed throughout the sample period. The second row requires these classifications to remain fixed across the pre- and post-reform periods. Counts are independently rounded in accordance with the disclosure requirements applied to the administrative data and may therefore not sum exactly.

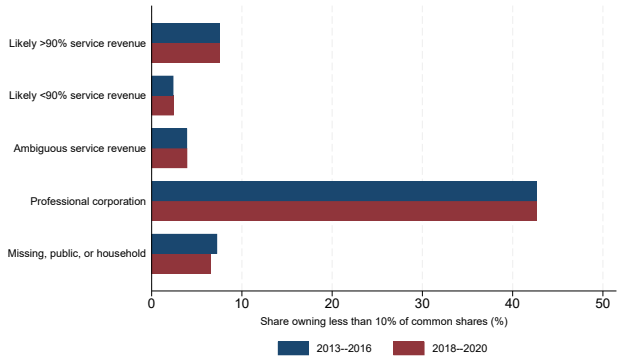
A.4 Changes Along the Observable TOSI Exemption Margins

Figure A1 examines whether spouses adjusted their common-share ownership or employment in response to the TOSI reform. The sample consists of spouses eligible to receive dividends, and exposure is defined only by the age-based rules. The sample therefore does not condition on spouses remaining on the same side of the ownership or employment criteria used in the main treatment classification.

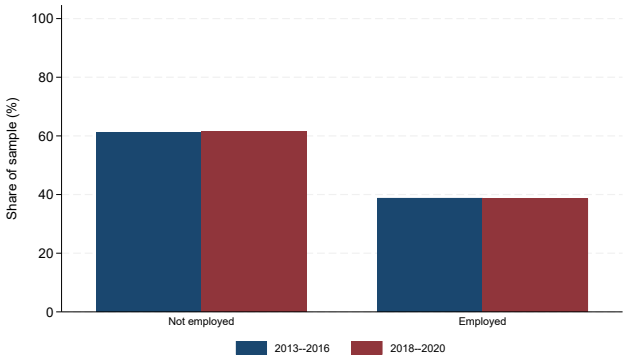
Figure A1: Changes Along the Observable TOSI Exemption Margins



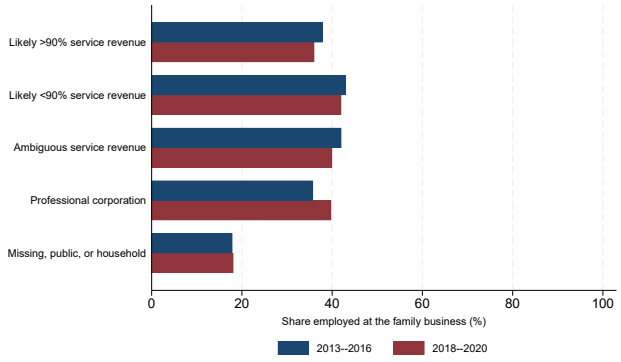
(a) Common-share ownership



(b) Common-share ownership by business category



(c) Employment in the family corporation



(d) Employment by business category

Notes: The sample consists of spouses eligible to receive dividends. Age-exposed and age-exempt spouses are classified using only the age-based rules and without conditioning on their ownership or employment choices. Panels A and B compare spouses' common-share ownership during 2013–2016 with ownership during 2018–2020. Panel A reports the shares below and at or above the 10% threshold. Panel B reports the share below the threshold by business category. Panels C and D compare employment in the family corporation during 2013–2016 with employment during 2018–2020, overall and by business category. All comparisons are descriptive.

A.5 Alternative Exposure Definitions

The two alternative definitions retain the primary owner's age restriction but change how spouses are classified. A3 compares spouses whose common-share ownership remains below 10% with spouses whose ownership remains at or above 10%, without using the employment criterion. A3B makes the same comparison among spouse-year observations with no recorded employment in the family corporation. Under both definitions, spouse dividend income and dividend receipt decline after implementation, while primary-owner dividend income rises. The estimated decline in log spouse dividends is similar to the main estimate under A3 and somewhat smaller under A3B. The positive estimate for family tax liability in the main specification is not statistically significant under either alternative. Because employment in A3B is observed each year and may itself respond to the reform, I treat that specification as a sensitivity check.

Table A4: Income Responses under Alternative Exposure Definitions

	A3			A3B		
	2017	2018–2020	Observations	2017	2018–2020	Observations
Panel A. Spouse						
Dividend income, log	0.091***	-0.735***	248,530	0.127***	-0.624***	112,655
	(0.025)	(0.049)		(0.035)	(0.067)	
Dividend income, dollars	5,244***	-14,201***	492,680	3,401***	-10,723***	226,090
	(586)	(711)		(707)	(848)	
Dividend income, PPML	0.118***	-0.899***	365,535	0.099***	-1.002***	163,585
	(0.016)	(0.037)		(0.023)	(0.056)	
Dividend receipt	0.014***	-0.159***	492,680	0.004	-0.128***	226,090
	(0.005)	(0.007)		(0.007)	(0.009)	
T4 income, log	-0.038***	-0.026*	389,025	-0.047**	-0.053***	156,615
	(0.014)	(0.015)		(0.020)	(0.020)	
T4 income, dollars	-893	-3,865***	492,680	-27	-3,706***	226,090
	(1,272)	(809)		(2,150)	(1,261)	
Family-business T4, log	0.003	0.029	222,280	—	—	—
	(0.017)	(0.020)				
Family-business T4, dollars	-1,640**	-4,100***	222,475	—	—	—
	(704)	(901)				
Panel B. Primary owner						
Dividend income, log	-0.030	0.196***	265,565	-0.009	0.113***	127,270
	(0.028)	(0.033)		(0.036)	(0.042)	
Dividend income, dollars	421	9,521***	492,680	1,295	5,500***	226,090
	(1,611)	(990)		(2,690)	(1,126)	
Dividend income, PPML	-0.018	0.189***	376,975	0.001	0.102***	174,615
	(0.038)	(0.024)		(0.055)	(0.029)	
Dividend receipt	-0.014**	0.040***	492,680	-0.006	0.035***	226,090
	(0.006)	(0.006)		(0.007)	(0.008)	
T4 income, log	0.030**	0.105***	386,295	0.024	0.108***	153,380
	(0.014)	(0.016)		(0.021)	(0.023)	
T4 income, dollars	1,588*	8,887***	492,680	1,953	8,591***	226,090
	(901)	(1,316)		(1,190)	(1,483)	
Family-business T4, log	-0.005	0.030	183,055	—	—	—
	(0.017)	(0.021)				
Family-business T4, dollars	762	8,085***	183,245	—	—	—
	(1,209)	(1,714)				
Panel C. Couple						
Taxable income, log	0.006	-0.011*	492,205	-0.004	-0.013*	225,865
	(0.006)	(0.006)		(0.007)	(0.007)	
Taxable income, dollars	5,668**	1,405	492,680	3,596	-1,646	226,090
	(2,475)	(1,985)		(3,896)	(1,939)	
Tax liability, log	-0.008	0.018	469,850	-0.026	-0.002	217,740
	(0.013)	(0.014)		(0.017)	(0.018)	
Tax liability, dollars	934	1,508*	492,680	143	-464	226,090
	(979)	(900)		(1,552)	(812)	

Notes: Columns report treated interactions with 2017 or Post; Post covers 2018–2020. A3 compares spouses whose common-share ownership remains below 10% with spouses whose ownership remains at or above 10%, retaining the age restriction and without using employment to classify exposure. A3B applies this comparison only to spouse-year observations with no recorded employment at the family corporation; family-business T4 outcomes are omitted (—). Dollar outcomes are in dollars; receipt is a zero-one indicator. PPML uses dividend income in levels and retains zero outcomes. Standard errors are in parentheses. Observation counts are reported separately for each outcome and definition. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

A.6 Unbalanced Panel

The dividend-income and receipt estimates remain close to the balanced-panel results. The spouse family-business T4 estimate is smaller and no longer statistically clear in the unbalanced panel. Table A5 reports the comparison.

Table A5: Income Responses in the Balanced and Unbalanced Panels

	Balanced panel	Unbalanced panel
A. Dividends		
Spouse dividend income	-0.753***	-0.739***
	(0.066)	(0.047)
Observations	239,795	414,920
Primary-owner dividend income	0.134***	0.124***
	(0.037)	(0.029)
Observations	255,870	451,240
Spouse receives dividends	-0.168***	-0.168***
	(0.009)	(0.006)
Observations	478,735	904,315
Primary owner receives dividends	0.029***	0.032***
	(0.007)	(0.006)
Observations	478,735	904,315
B. Employment income		
Spouse T4 income	-0.079***	-0.088***
	(0.020)	(0.014)
Observations	377,555	692,140
Primary-owner T4 income	0.153***	0.143***
	(0.024)	(0.017)
Observations	374,695	683,610
Spouse T4 from family business	-0.133***	-0.062
	(0.047)	(0.041)
Observations	215,465	356,555
Primary-owner T4 from family business	-0.040	-0.051
	(0.051)	(0.054)
Observations	177,420	288,940
C. Family outcomes		
Family taxable income	-0.012*	-0.006
	(0.007)	(0.005)
Observations	478,260	902,805
Family tax liability	0.032*	0.027**
	(0.017)	(0.012)
Observations	456,090	853,635

Notes: TC implementation-period estimates for 2018–2020. Income outcomes are in logs and dividend receipt is a zero-one indicator. Family-clustered standard errors are in parentheses. Observations are rounded as released. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

A.7 Simulated Tax Penalty and Quartile Rankings

To rank treated families by the potential tax cost of continuing to split income, I hold both spouses' observed 2016 income components fixed and calculate total personal tax liability under two tax systems. The first applies the expanded TOSI rules to the spouse's dividend income, while the second taxes that income under the ordinary personal income-tax system. Both calculations use the same year-specific federal and provincial tax parameters, and neither reassigns income between spouses. I define the dollar penalty as

$$Penalty_{it} = Treated_i \left[Tax_t^{TOSI} \left(FamInc_{i,2016} \right) - Tax_t^{NoTOSI} \left(FamInc_{i,2016} \right) \right], \tag{7}$$

where $FamInc_{i,2016}$ denotes the observed 2016 income components of the spouse and primary owner. The measure is zero before implementation and for control families in every year. From 2018 onward, it varies across treated families with the additional tax applied to the spouse's fixed dividend income. I rank treated families by their average simulated dollar penalty over 2018–2020 and hold the resulting quartile assignments fixed. Each penalty-quartile DiD regression compares the relevant treated group with the same control group. The tax-incentive IV uses the same ranking when examining responses by penalty quartile. The ranking therefore measures the potential cost of maintaining the pre-reform income allocation, rather than subsequent income responses.

B Additional Difference-in-Differences Results

This appendix reports supporting estimates for the difference-in-differences analysis in Section 4. Dollar estimates complement the log-income results, while PPML provides an additional dividend specification that retains zero outcomes. The quartile tables report the full numerical estimates underlying the main figures.

B.1 Estimates in Dollars and PPML

B.2 Dividend Responses by Family Income

Table B1: Difference-in-Differences Estimates in Dollars and PPML

Panel A. Dividend income, 2018–2020

	Dividend income ($)			PPML dividend income	
	Spouse	Primary owner	Couple	Spouse	Primary owner
Treated × Post	-12,391***	7,124***	-5,267***	-1.061***	0.125***
	(815)	(1,149)	(1,313)	(0.052)	(0.026)
Observations	478,735	478,735	478,735	353,100	364,340

Panel B. Labour income ($)

	2017	2018–2020
Spouse, all employers	278	-4,419***
	(2,071)	(1,136)
Primary owner, all employers	2,442**	10,867***
	(1,138)	(1,505)
Spouse, family corporation	-1,846*	-11,503***
	(1,063)	(1,296)
Primary owner, family corporation	-3,222	9,729*
	(3,447)	(5,174)

Panel C. Family taxable income and tax liability ($), 2018–2020

	Family taxable income	Family tax liability
Treated × Post	928	225
	(1,875)	(755)
Observations	478,735	478,735

Notes: All panels report difference-in-differences estimates comparing treated and control families. Post covers 2018–2020. Dollar coefficients and standard errors are measured in dollars. PPML includes zero dividend outcomes; its coefficients are on the log-link scale. The labour-income rows report the treated interaction with 2017 or Post. Standard errors clustered at the family level appear in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table B2: Dividend Responses by 2016 Family-Income Quartile

	Q1	Q2	Q3	Q4
Panel A. Log dividends				
Spouse dividend income	-0.041	-0.356***	-0.657***	-0.975***
	(0.236)	(0.128)	(0.116)	(0.101)
Observations	32,775	62,030	72,345	70,650
Primary-owner dividend income	0.097	0.013	0.082	0.244***
	(0.124)	(0.075)	(0.066)	(0.058)
Observations	35,120	66,110	76,855	75,765
Panel B. Dividends in dollars				
Spouse dividends	-2,015***	-4,569***	-8,315***	-20,099***
	(715)	(810)	(910)	(1,561)
Primary-owner dividends	187	500	2,400	13,112***
	(1,185)	(1,363)	(1,666)	(2,175)

Notes: Quartiles use family income in 2016. All coefficients are difference-in-differences estimates comparing treated and control families and cover 2018–2020. Panel A reports log-dividend coefficients; Panel B reports coefficients and standard errors in dollars. Log and dollar specifications can use different samples. Standard errors clustered at the family level are in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

B.3 Labour Income Responses

The employment-income estimates distinguish total T4 income from wages paid by the linked family corporation. In the dollar specification, spouses' total T4 income falls by $4,419 and owners' rises by $10,867, while business-paid spouse T4 income falls by $11,503 (Panel B of Table B1). Treated and control spouses followed different trends in total T4 income before the reform, so that decline is harder to attribute to TOSI than the decline in wages paid by the family business. The smaller estimated fall in total than in business-paid spouse T4 income is consistent with earnings from other employers providing an offset, but the outcomes use different samples and do not measure that offset directly. Main Table 2 reports the log estimates, and Panels C and D of Figure 4 show their annual paths.

Table B3 reports the log and dollar estimates by family-income quartile. Owners' total T4 income rises across the distribution, while the clearest decline in spouses' total T4 income occurs in Q4. These estimates describe changes in reported employment income and do not establish changes in hours worked.

Table B3: Treatment–Control Labour Income Responses by 2016 Family-Income Quartile

	Q1	Q2	Q3	Q4
A. Log income				
Spouse, all employers	0.150	0.042	-0.043	-0.146***
	(0.130)	(0.050)	(0.035)	(0.027)
Observations	75,695	104,500	106,340	88,640
Primary owner, all employers	0.177*	0.183***	0.150***	0.093***
	(0.091)	(0.063)	(0.045)	(0.032)
Observations	75,415	103,280	104,845	88,780
Spouse, family corporation	0.178	0.033	0.022	-0.304***
	(0.127)	(0.102)	(0.089)	(0.070)
Observations	52,680	62,700	55,105	42,995
Primary owner, family corporation	0.035	-0.166	-0.004	-0.037
	(0.099)	(0.175)	(0.114)	(0.062)
Observations	45,775	52,145	43,520	34,175
B. Income in dollars				
Spouse, all employers	2,027**	578	-1,721*	-9,535***
	(1,026)	(873)	(985)	(2,389)
Observations	104,425	132,450	131,085	108,060
Primary owner, all employers	966	3,050*	4,480***	11,058***
	(2,163)	(1,742)	(1,720)	(2,927)
Observations	104,425	132,450	131,085	108,060
Spouse, family corporation	-555	-3,039***	-4,021**	-21,318***
	(1,351)	(1,017)	(1,804)	(2,497)
Observations	52,740	62,770	55,140	43,030
Primary owner, family corporation	-342	-3,852*	389	18,423**
	(2,445)	(2,162)	(4,736)	(8,579)
Observations	45,850	52,195	43,550	34,220

Notes: Implementation-period estimates for 2018–2020. Income quartiles are fixed using 2016 family income. Standard errors clustered at the family level are in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$. A dash indicates an unreported specification.

B.4 Family Taxable Income and Tax Liability by Family Income

Table B4: Family Taxable Income and Tax Liability Responses by 2016 Family-Income Quartile

	Q1	Q2	Q3	Q4
Panel A. Log outcomes				
Family taxable income	-0.007	-0.019	-0.019	-0.005
	(0.040)	(0.018)	(0.012)	(0.010)
Family tax liability	0.205	0.066	0.035	0.021
	(0.152)	(0.047)	(0.026)	(0.018)
Panel B. Dollar outcomes				
Family taxable income	533	-4,211*	-4,656*	-4,552
	(2,851)	(2,544)	(2,380)	(3,909)
Family tax liability	162	-819	-374	-1,918
	(766)	(792)	(695)	(1,612)

Notes: Quartiles use family income in 2016. All coefficients are difference-in-differences estimates comparing treated and control families and cover 2018–2020. Panel A reports log-outcome coefficients; Panel B reports coefficients and standard errors in dollars. Standard errors clustered at the family level are in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table B5: Difference-in-Differences Estimates of Corporate Outcomes

Panel A. Announcement and implementation estimates

	Log retained earnings	Log total assets	Annual asset growth	Annual retained-earnings growth
Announcement, 2017	0.025	0.042***	-1,069.739	4.458
	(0.017)	(0.015)	(951.680)	(63.405)
Implementation, 2018–2020	0.018	0.037*	386.884	2.059
	(0.020)	(0.020)	(699.107)	(58.462)
Observations	344,070	474,320	423,970	419,630

Panel B. Event-study estimates for retained earnings and assets

	Log retained earnings	Log total assets
2013	-0.115***	-0.103***
	(0.024)	(0.022)
2014	-0.063***	-0.052***
	(0.019)	(0.016)
2015	-0.005	-0.024**
	(0.013)	(0.010)
2017	-0.016	-0.000
	(0.013)	(0.011)
2018	-0.003	0.021
	(0.017)	(0.014)
2019	-0.016	0.006
	(0.020)	(0.018)
2020	-0.048**	-0.040*
	(0.023)	(0.024)
Observations	344,070	474,320

Notes: Panel A reports the treatment interactions with the 2017 announcement indicator and the 2018–2020 implementation indicator from the same DiD regression. Panel B reports annual treatment interactions relative to the omitted year 2016. Retained earnings are measured as the average of beginning- and end-of-year retained earnings before taking logs. Annual growth equals 100 times the change from the previous year divided by the previous-year value, so growth coefficients are in percentage points. Each outcome has its own estimation sample. Specifications include the controls and fixed effects described in Section 4.1. Family-clustered standard errors are in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

B.5 Responses by Simulated Tax-Penalty Quartile

The simulated dollar penalty used to form the quartiles is defined in Appendix A.7. The following tables report log-income, dividend-receipt, and dollar responses for all four quartiles, including outcomes summarized in the main figures.

Table B6: Difference-in-Differences Responses by Simulated TOSI-Penalty Quartile

	Q1	Q2	Q3	Q4
Panel A. Dividend income and receipt				
Spouse dividend income	0.020	0.260**	-0.895***	-1.374***
	(0.215)	(0.110)	(0.110)	(0.111)
Observations	228,505	229,735	231,555	233,145
Primary-owner dividend income	0.073	0.083	0.100	0.256***
	(0.094)	(0.073)	(0.063)	(0.056)
Observations	242,160	242,975	243,855	245,500
Spouse receives dividends	0.031**	0.037***	-0.297***	-0.381***
	(0.015)	(0.014)	(0.019)	(0.016)
Observations	460,780	462,080	461,720	463,535
Primary owner receives dividends	0.030*	0.044***	0.013	0.037***
	(0.016)	(0.013)	(0.014)	(0.012)
Observations	460,780	462,080	461,720	463,535
Panel B. Employment income				
Spouse T4 income	-0.091***	-0.085***	-0.177***	0.027
	(0.034)	(0.028)	(0.039)	(0.046)
Observations	365,115	366,220	365,570	365,260
Primary-owner T4 income	0.075	0.043	0.239***	0.281***
	(0.047)	(0.039)	(0.051)	(0.049)
Observations	362,405	363,385	363,040	363,920
Spouse T4 from family business	-0.283	0.010	-0.159*	-0.135**
	(0.199)	(0.112)	(0.097)	(0.061)
Observations	214,005	214,145	214,365	214,875
Primary-owner T4 from family business	-0.305	0.086	-0.208	0.026
	(0.203)	(0.077)	(0.153)	(0.058)
Observations	176,050	176,120	176,420	176,910
Panel C. Family taxable income and tax liability				
Family taxable income	0.040***	0.024**	-0.013	-0.059***
	(0.014)	(0.012)	(0.015)	(0.012)
Observations	460,310	461,610	461,250	463,065
Family tax liability	0.009	-0.042	0.028	0.150***
	(0.033)	(0.030)	(0.034)	(0.027)
Observations	438,395	439,725	439,375	441,175

Notes: Treated families are ranked by their average simulated 2018–2020 dollar penalty calculated using fixed 2016 income. Each quartile is compared separately with the same control group. Coefficients cover 2018–2020. Income and tax outcomes are in logs; dividend receipt is a zero-one indicator. Observation counts are rounded as released. Family-clustered standard errors are in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table B7: Dollar Responses by Simulated TOSI-Penalty Quartile

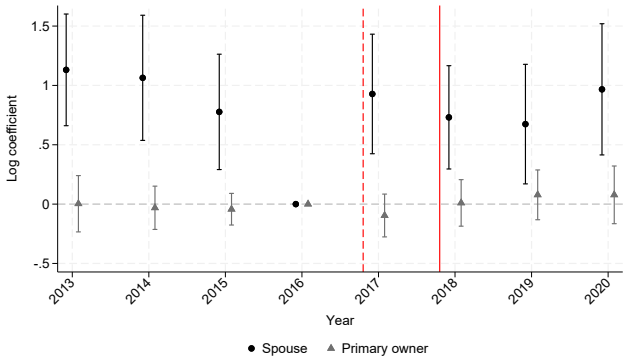
	Q1	Q2	Q3	Q4
Spouse dividends	2.93	2.83	-6.16	-40.76
	(0.68)	(0.69)	(1.07)	(1.79)
Primary-owner dividends	2.60	1.92	2.52	19.82
	(1.96)	(1.42)	(2.20)	(2.51)
Primary-owner total T4 income	4.96	0.14	10.23	28.60
	(2.37)	(2.09)	(3.53)	(3.19)
Family taxable income	12.80	5.31	-2.30	-3.02
	(3.06)	(2.79)	(4.20)	(3.38)
Family tax liability	0.72	-1.07	-3.80	4.84
	(0.97)	(1.24)	(1.76)	(1.32)

Notes: Entries are treatment–control implementation-period estimates for 2018–2020, in thousands of dollars; family-clustered standard errors are in parentheses. Treated families are ranked by their average simulated penalty for 2018–2020 using fixed 2016 income, and each quartile is compared separately with the same control group. Outcome samples can differ. Log specifications appear in Appendix Table B6.

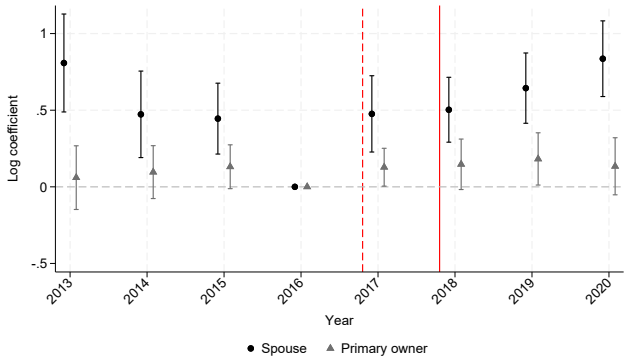
B.6 Event Studies by Simulated Tax-Penalty Quartile

These figures report annual coefficients for the penalty-ranked DiD comparisons. They supplement the post-implementation coefficients by showing pre-reform paths, announcement responses, and the timing of subsequent adjustments.

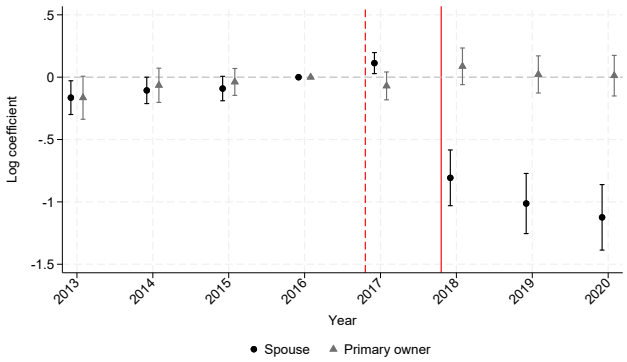
Figure B1: Event-Study Estimates of Dividend Income by Simulated TOSI-Penalty Quartile



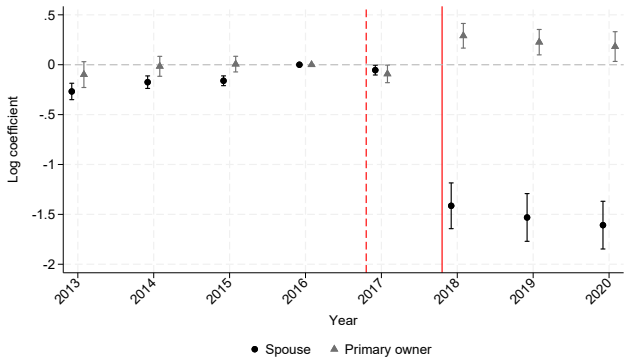
(a) Penalty quartile 1



(b) Penalty quartile 2



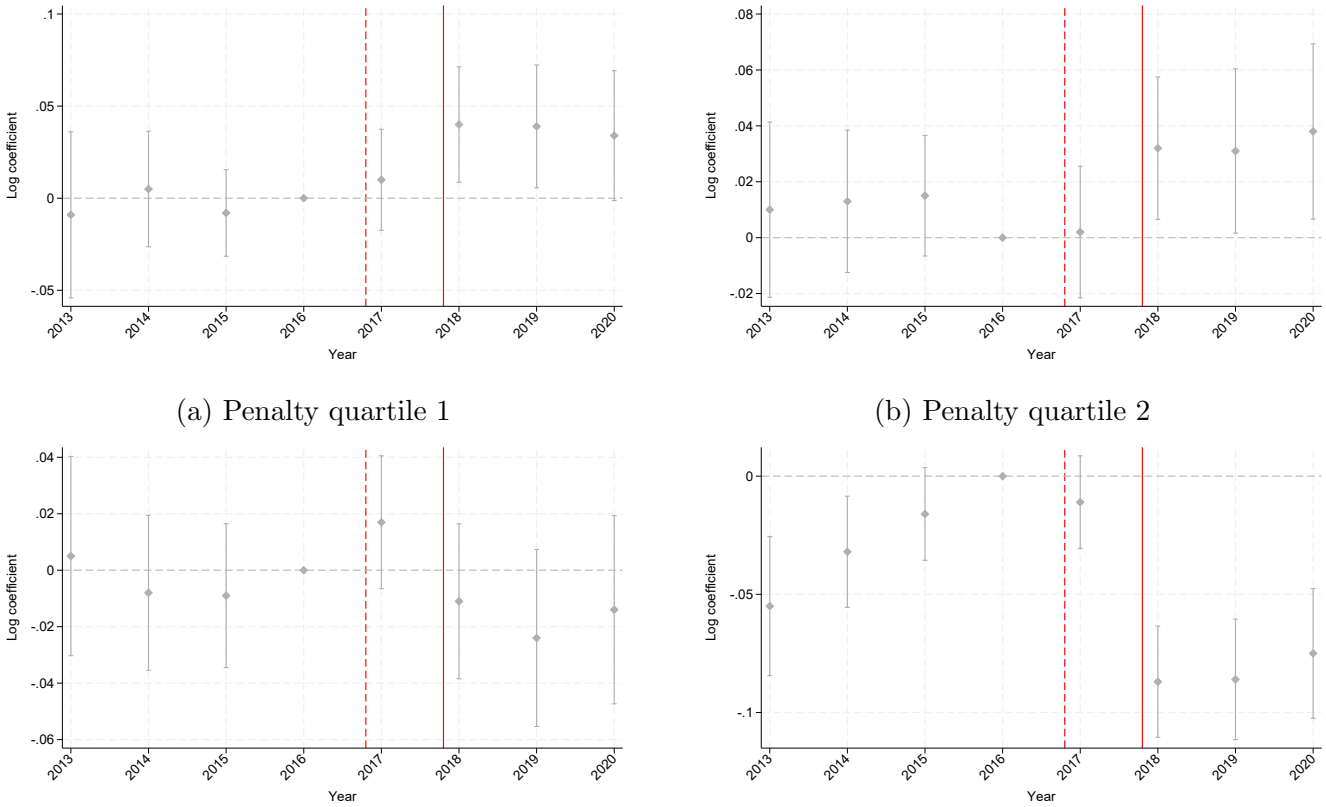
(c) Penalty quartile 3



(d) Penalty quartile 4

Notes: Each panel compares one quartile of treated families, ranked by their average simulated 2018–2020 TOSI penalty using fixed 2016 income, with the same control group. Estimates are relative to 2016. Dots show treatment–control event-study coefficients and bars show 95% confidence intervals based on family-clustered standard errors. The dashed line separates the pre-reform years from the 2017 announcement year; the solid line separates the announcement year from implementation in 2018.

Figure B2: Event-Study Estimates of Family Taxable Income by Simulated TOSI-Penalty Quartile



(a) Penalty quartile 1

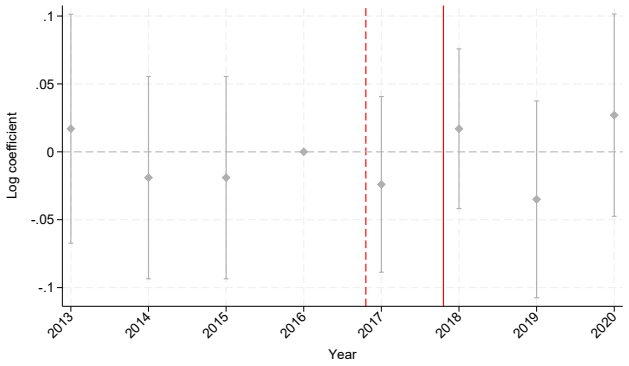
(b) Penalty quartile 2

(c) Penalty quartile 3

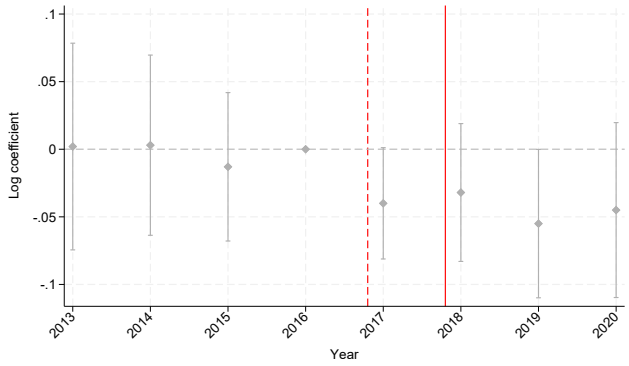
(d) Penalty quartile 4

Notes: Each panel compares one quartile of treated families, ranked by their average simulated 2018–2020 TOSI penalty using fixed 2016 income, with the same control group. Estimates are relative to 2016. Dots show treatment–control event-study coefficients and bars show 95% confidence intervals based on family-clustered standard errors. The dashed line separates the pre-reform years from the 2017 announcement year; the solid line separates the announcement year from implementation in 2018.

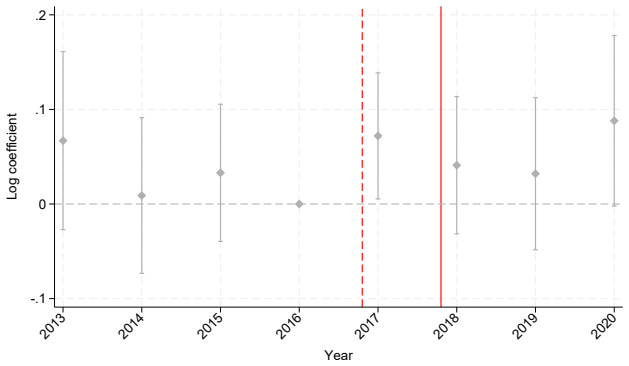
Figure B3: Event-Study Estimates of Family Tax Liability by Simulated TOSI-Penalty Quartile



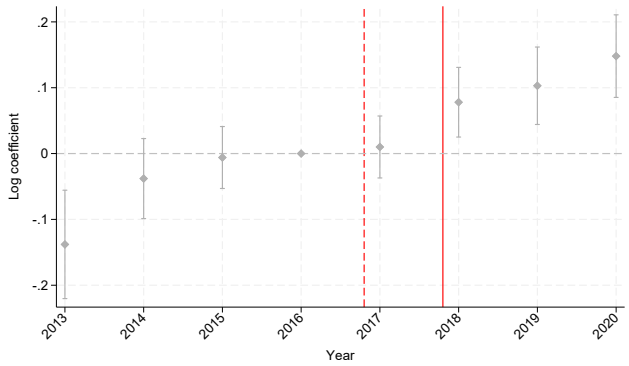
(a) Penalty quartile 1



(b) Penalty quartile 2



(c) Penalty quartile 3



(d) Penalty quartile 4

Notes: Each panel compares one quartile of treated families, ranked by their average simulated 2018–2020 TOSI penalty using fixed 2016 income, with the same control group. Estimates are relative to 2016. Dots show treatment–control event-study coefficients and bars show 95% confidence intervals based on family-clustered standard errors. The dashed line separates the pre-reform years from the 2017 announcement year; the solid line separates the announcement year from implementation in 2018.

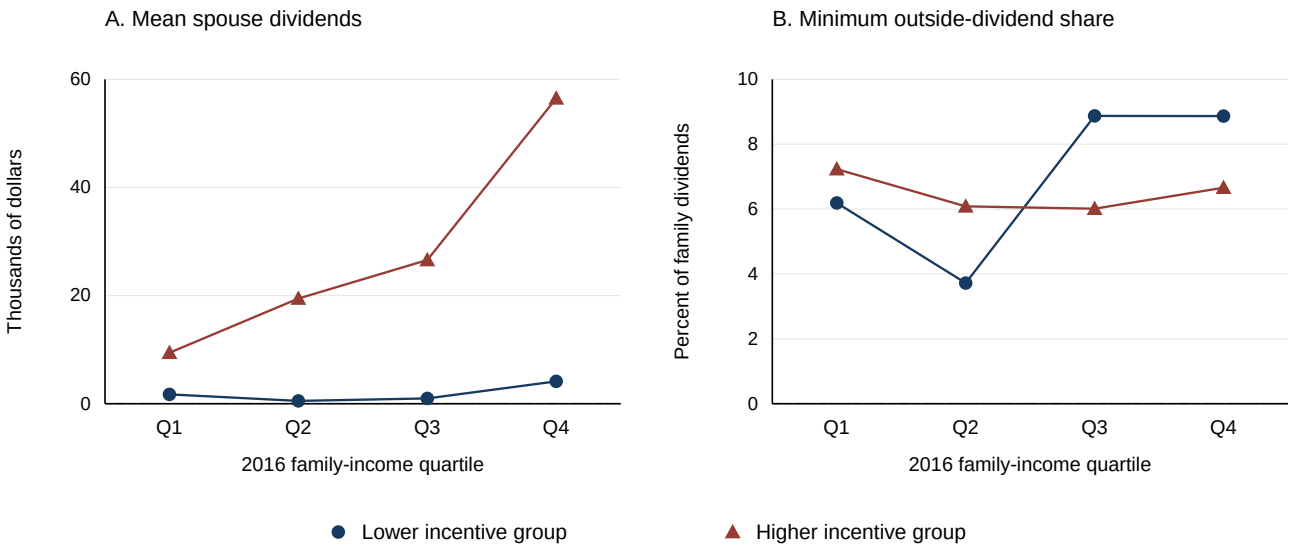
C Tax-Incentive IV Estimates

This appendix reports supporting estimates for the contemporaneous tax-incentive IV in Section 5. These estimates instrument the actual log spouse-to-owner net-of-tax-rate ratio with a simulated ratio calculated using fixed 2016 incomes.

C.1 Pre-Reform Dividends and Dividend Sources

Spouses in the higher potential-incentive group receive substantially larger dividends at each family-income rank. The mean minimum share of family dividends from outside the linked corporation ranges from 3.7% to 8.9%, with no uniform increase across incentive groups (Figure C1). Different proportions of unaffected outside dividends could dilute the measured response of total spouse dividends. This comparison provides descriptive evidence about that concern, while the family-level lower bound cannot establish each spouse's actual dividend sources. The potential incentive used for this descriptive comparison is calculated before adding dividends to the spouse and does not reallocate them to the owner.

Figure C1: Pre-Reform Dividends and Dividend Sources by Income and Incentive Group



Notes: The figure reports released 2016 cell means for treated families, grouped by the family-income quartiles used in the regressions and the two pre-reform incentive groups. Panel A reports cash dividends received by the spouse. Panel B reports the mean minimum outside-dividend share, calculated as the positive part of family dividends minus total dividends paid by the linked corporation, divided by family dividends. This is a family-level lower bound and does not identify the spouse's actual outside-dividend share. The potential descriptive incentive compares the spouse's net-of-dividend-tax rate with spouse dividends removed with the owner's rate at observed income, without reallocating the spouse's dividends to the owner. It differs from the incentive instrumented in the main IV. No confidence intervals are reported because the release supplies cell means rather than sampling uncertainty.

C.2 Additional Outcomes for the Full Sample

Table C1 reports outcomes beyond the core dividend and family-income measures in the main table. Total owner T4 income responds as the incentive falls, while wages paid by the family corporation do not change clearly. Owner tax liability also rises as the incentive falls, whereas the family-tax estimate remains imprecise. These outcomes describe different estimation samples and do not supply an exact accounting of the same dollars. Corporate asset and retained-earnings estimates are statistically insignificant for the full sample.

C.3 Dividend Responses by Family Income

C.4 Family Income and Tax Responses by Family Income

C.5 Employment-Income Responses by Family Income

C.6 Corporate Responses by Family Income

C.7 Responses by Simulated-Penalty Quartile

The penalty-ranked IV estimates measure responses to the relative tax incentive within the released penalty groups, rather than the total effect of belonging to each treated group. Spouse-dividend elasticities are positive but do not increase monotonically with the penalty rank. The family-income and tax estimates are generally imprecise. In Q1, positive coefficients on family taxable income and tax liability imply declines as the incentive falls; these are distinct from the penalty-ranked DiD treatment effects. The estimates do not establish a systematic gradient in family-level responses.

C.8 Business Structure and Alternative Definitions

Both professional and non-professional families show positive spouse-dividend elasticities, while owner-dividend adjustment is clearer among professionals. For multiple-business families, receipt and the spouse's dividend share respond, but the log-dividend amount is imprecise. Table C7 reports these comparisons. The unbalanced sample and alternative treatment definitions preserve the direction of the main dividend responses (Table C8). The spouse elasticity is 1.385 in the unbalanced sample, 2.278 under A3, 1.212 under A3B, and 1.158 under E3. A3 therefore gives a

Table C1: Additional Outcomes in the Tax-Incentive IV

Outcome	S_{it}	Observations [F-statistic]
	-0.041*	478,540
Owner dividend receipt	(0.024)	[2,932.00]
First stage	0.450*** (0.008)	
	-0.007	377,475
Log spouse total T4 income	(0.064)	[2,010.00]
First stage	0.421*** (0.009)	
	-0.138**	374,595
Log owner total T4 income	(0.063)	[1,981.00]
First stage	0.437*** (0.010)	
	0.028	215,190
Log spouse business-paid T4 income	(0.138)	[320.40]
First stage	0.215*** (0.012)	
	0.135	177,055
Log owner business-paid T4 income	(0.152)	[261.40]
First stage	0.210*** (0.013)	
	0.032	343,975
Log retained earnings	(0.069)	[2,067.00]
First stage	0.459*** (0.010)	
	0.059	474,135
Log assets	(0.064)	[2,882.00]
First stage	0.450*** (0.008)	
	-4,648.750	423,735
Annual asset growth	(4,950.144)	[2,577.00]
First stage	0.466*** (0.009)	
	59.752	419,335
Annual retained-earnings growth	(427.376)	[2,568.00]
First stage	0.467*** (0.009)	
	0.081	424,030
Log spouse tax liability	(0.089)	[2,469.00]
First stage	0.432*** (0.009)	
	-0.301***	430,000
Log owner tax liability	(0.079)	[2,890.00]
First stage	0.472*** (0.009)	

Notes: The shifting incentive S_{it} and simulated instrument follow Equations (3)–(6). All specifications retain the same controls and fixed effects. First stages use S_{it} as their dependent variable and the sample of the corresponding outcome. Family-clustered standard errors are in parentheses; first-stage F-statistics are below observations. Receipt is a zero-one indicator. Annual-growth outcomes retain their released units. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table C2: Dividend Responses by Family-Income Quartile in the Tax-Incentive IV

	Q1	Q2	Q3	Q4
Log spouse dividends				
	1.222*	1.135***	1.692***	1.282***
S_{it}	(0.674)	(0.364)	(0.392)	(0.312)
First stage	0.244*** (0.027)	0.367*** (0.022)	0.369*** (0.027)	0.496*** (0.024)
Observations	32,740	62,010	72,330	70,650
[F-statistic]	[83.73]	[281.70]	[189.40]	[430.60]
Log owner dividends				
	0.435	-0.035	-0.327	-0.739***
S_{it}	(0.415)	(0.228)	(0.218)	(0.212)
First stage	0.344*** (0.028)	0.460*** (0.020)	0.504*** (0.025)	0.592*** (0.022)
Observations	35,065	66,080	76,845	75,765
[F-statistic]	[155.80]	[524.70]	[404.10]	[719.50]
Spouse dividend receipt				
	0.023	0.242***	0.341***	0.246***
S_{it}	(0.068)	(0.055)	(0.053)	(0.044)
First stage	0.258*** (0.015)	0.447*** (0.015)	0.496*** (0.020)	0.586*** (0.016)
Observations	104,295	132,400	131,065	108,055
[F-statistic]	[288.70]	[945.10]	[602.50]	[1,309.00]
Spouse share of couple dividends				
	3.971	16.113***	25.474***	29.489***
S_{it}	(6.725)	(4.275)	(4.517)	(4.308)
First stage	0.331*** (0.025)	0.457*** (0.019)	0.480*** (0.024)	0.577*** (0.020)
Observations	41,855	75,735	88,635	86,575
[F-statistic]	[176.50]	[606.00]	[407.40]	[872.70]

Notes: Both stages follow Equations (5) and (6). The shifting incentive S_{it} is defined in Equation (3) and is the first-stage dependent variable. Each second-stage outcome has its own sample and first stage. Family-clustered standard errors are in parentheses; first-stage F-statistics are in square brackets below observations. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$. Income quartiles use total family income in 2016 and are fixed throughout the analysis. The lowest quartile is retained. Dividend receipt is a zero-one indicator and the spouse dividend share is in percentage points.

Table C3: Family Income and Tax Responses by Family-Income Quartile in the Tax-Incentive IV

	Q1	Q2	Q3	Q4
Log family taxable income				
	0.045	0.032	0.045	-0.030
S_{it}	(0.102)	(0.053)	(0.038)	(0.041)
First stage	0.258*** (0.015)	0.447*** (0.015)	0.496*** (0.020)	0.586*** (0.016)
Observations	103,915	132,360	131,040	108,035
[F-statistic]	[288.50]	[945.10]	[602.60]	[1,309.00]
Log family tax liability				
	-0.106	-0.015	-0.152*	-0.040
S_{it}	(0.385)	(0.133)	(0.086)	(0.070)
First stage	0.259*** (0.017)	0.448*** (0.015)	0.498*** (0.020)	0.587*** (0.016)
Observations	85,940	129,700	130,045	107,640
[F-statistic]	[236.20]	[940.30]	[601.60]	[1,316.00]

Notes: Both stages follow Equations (5) and (6). The shifting incentive S_{it} is defined in Equation (3) and is the first-stage dependent variable. Each second-stage outcome has its own sample and first stage. Family-clustered standard errors are in parentheses; first-stage F-statistics are in square brackets below observations. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$. Income quartiles use total family income in 2016 and are fixed throughout the analysis. The lowest quartile is retained.

larger estimate rather than numerical invariance. Family tax liability responds significantly in the unbalanced and A3 specifications, while it is imprecise under the main, A3B, and E3 definitions.

Table C4: Employment-Income Responses by Family-Income Quartile in the Tax-Incentive IV

	Q1	Q2	Q3	Q4
Log spouse total T4 income				
	-0.307	-0.199	0.106	0.085
S_{it}	(0.255)	(0.140)	(0.115)	(0.091)
First stage	0.215*** (0.017)	0.409*** (0.017)	0.465*** (0.023)	0.567*** (0.015)
Observations	75,645	104,475	106,330	88,635
[F-statistic]	[154.80]	[563.00]	[398.20]	[1,419.00]
Log owner total T4 income				
	-0.186	-0.490***	-0.146	0.104
S_{it}	(0.207)	(0.141)	(0.128)	(0.091)
First stage	0.237*** (0.018)	0.417*** (0.018)	0.472*** (0.024)	0.584*** (0.018)
Observations	75,350	103,250	104,835	88,780
[F-statistic]	[178.70]	[564.40]	[387.50]	[1,002.00]
Log spouse business-paid T4 income				
	-0.021	-0.174	-0.009	0.066
S_{it}	(0.405)	(0.357)	(0.393)	(0.166)
First stage	0.115*** (0.018)	0.190*** (0.022)	0.165*** (0.025)	0.422*** (0.028)
Observations	52,660	62,685	55,060	42,960
[F-statistic]	[39.42]	[76.89]	[44.70]	[231.90]
Log owner business-paid T4 income				
	0.341	-0.068	-0.093	0.053
S_{it}	(0.414)	(0.426)	(0.401)	(0.172)
First stage	0.125*** (0.020)	0.179*** (0.023)	0.182*** (0.028)	0.397*** (0.031)
Observations	45,770	52,130	43,370	34,115
[F-statistic]	[39.90]	[61.00]	[43.13]	[167.60]

Notes: Both stages follow Equations (5) and (6). The shifting incentive S_{it} is defined in Equation (3) and is the first-stage dependent variable. Each second-stage outcome has its own sample and first stage. Family-clustered standard errors are in parentheses; first-stage F-statistics are in square brackets below observations. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$. Income quartiles use total family income in 2016 and are fixed throughout the analysis. The lowest quartile is retained.

Table C5: Corporate Responses by Family-Income Quartile in the Tax-Incentive IV

	Q1	Q2	Q3	Q4
Log retained earnings				
	0.738**	-0.149	-0.061	-0.031
S_{it}	(0.325)	(0.161)	(0.130)	(0.093)
First stage	0.235*** (0.019)	0.436*** (0.018)	0.478*** (0.024)	0.586*** (0.019)
Observations	58,130	92,660	100,805	90,140
[F-statistic]	[147.80]	[597.90]	[410.00]	[986.00]
Log assets				
	-0.021	0.074	-0.009	0.189*
S_{it}	(0.208)	(0.134)	(0.127)	(0.100)
First stage	0.258*** (0.015)	0.448*** (0.015)	0.493*** (0.020)	0.586*** (0.016)
Observations	103,200	131,230	129,900	107,090
[F-statistic]	[283.50]	[937.80]	[588.90]	[1,281.00]
Annual asset growth				
	-1,212.773	-355.613	-13,530.058	-2,906.123
S_{it}	(4,349.136)	(16,165.279)	(10,488.994)	(4,722.896)
First stage	0.259*** (0.016)	0.483*** (0.016)	0.519*** (0.023)	0.608*** (0.018)
Observations	92,005	117,175	116,090	95,905
[F-statistic]	[265.80]	[949.10]	[507.20]	[1,116.00]
Annual retained-earnings growth				
	-1,704.814	185.342	863.726	554.054
S_{it}	(1,656.904)	(443.529)	(807.378)	(808.591)
First stage	0.262*** (0.016)	0.484*** (0.016)	0.523*** (0.023)	0.608*** (0.018)
Observations	90,845	116,025	115,115	94,925
[F-statistic]	[271.40]	[955.00]	[499.40]	[1,112.00]

Notes: Both stages follow Equations (5) and (6). The shifting incentive S_{it} is defined in Equation (3) and is the first-stage dependent variable. Each second-stage outcome has its own sample and first stage. Family-clustered standard errors are in parentheses; first-stage F-statistics are in square brackets below observations. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$. Income quartiles use total family income in 2016 and are fixed throughout the analysis. The lowest quartile is retained. Annual growth outcomes retain the units of the released regressions.

Table C6: Tax-Incentive IV Responses by Simulated-Penalty Quartile

	Q1	Q2	Q3	Q4
Log spouse dividends				
	1.065***	0.627**	0.939***	0.468*
S_{it}	(0.365)	(0.316)	(0.307)	(0.245)
First stage	0.235*** (0.012)	0.255*** (0.013)	0.272*** (0.013)	0.301*** (0.013)
Observations	230,750	229,705	230,765	231,575
[F-statistic]	[372.10]	[410.40]	[458.60]	[562.60]
Log owner dividends				
	-0.190	-0.028	-0.098	-0.151
S_{it}	(0.247)	(0.203)	(0.225)	(0.225)
First stage	0.305*** (0.012)	0.347*** (0.013)	0.328*** (0.012)	0.317*** (0.012)
Observations	244,065	243,205	243,575	243,475
[F-statistic]	[619.10]	[744.00]	[701.40]	[686.90]
Spouse dividend receipt				
	0.157***	0.098***	0.169***	0.011
S_{it}	(0.046)	(0.038)	(0.046)	(0.047)
First stage	0.284*** (0.009)	0.324*** (0.009)	0.285*** (0.008)	0.264*** (0.008)
Observations	462,770	462,235	461,715	460,945
[F-statistic]	[1,100.00]	[1,349.00]	[1,138.00]	[1,049.00]
Spouse share of couple dividends				
	12.503***	3.126	12.767***	6.727*
S_{it}	(4.265)	(3.530)	(3.935)	(3.836)
First stage	0.304*** (0.012)	0.343*** (0.012)	0.328*** (0.012)	0.319*** (0.011)
Observations	281,930	280,960	281,385	281,215
[F-statistic]	[691.10]	[826.70]	[788.60]	[772.80]
Log family taxable income				
	0.132**	-0.056	-0.067	-0.038
S_{it}	(0.051)	(0.041)	(0.051)	(0.054)
First stage	0.284*** (0.009)	0.324*** (0.009)	0.286*** (0.008)	0.264*** (0.008)
Observations	462,300	461,765	461,245	460,475
[F-statistic]	[1,099.00]	[1,349.00]	[1,138.00]	[1,049.00]
Log family tax liability				
	0.259*	-0.032	-0.114	-0.120
S_{it}	(0.133)	(0.108)	(0.132)	(0.141)
First stage	0.292*** (0.009)	0.337*** (0.009)	0.295*** (0.009)	0.275*** (0.009)
Observations	440,430	439,905	439,360	438,640
[F-statistic]	[1,085.00]	[1,346.00]	[1,123.00]	[1,035.00]

Notes: Both stages follow Equations (5) and (6). The shifting incentive S_{it} is defined in Equation (3) and is the first-stage dependent variable. Each second-stage outcome has its own sample and first stage. Family-clustered standard errors are in parentheses; first-stage F-statistics are in square brackets below observations. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$. Quartiles refer to the fixed simulated-penalty ranking of treated families. Each estimation sample retains control families together with the relevant treated group. These coefficients measure responses to the relative tax incentive rather than total treatment effects. Receipt is a zero-one indicator and dividend share is in percentage points.

Table C7: Tax-Incentive IV Estimates by Business Structure

	Main	Professional	Non-professional	Multiple businesses
Log spouse dividends				
	1.299***	1.819***	0.826***	0.217
S_{it}	(0.193)	(0.492)	(0.218)	(0.491)
First stage	0.396*** (0.013)	0.481*** (0.033)	0.366*** (0.014)	0.421*** (0.021)
Observations	239,735	25,465	199,820	87,770
[F-statistic]	[975.70]	[215.80]	[657.20]	[396.70]
Log owner dividends				
	-0.342***	-1.256***	-0.045	-0.041
S_{it}	(0.118)	(0.263)	(0.149)	(0.351)
First stage	0.508*** (0.012)	0.639*** (0.030)	0.452*** (0.013)	0.467*** (0.020)
Observations	255,785	27,275	212,595	93,405
[F-statistic]	[1,927.00]	[455.20]	[1,195.00]	[542.50]
Spouse dividend receipt				
	0.243***	0.341***	0.186***	0.183**
S_{it}	(0.026)	(0.068)	(0.030)	(0.074)
First stage	0.450*** (0.008)	0.605*** (0.024)	0.405*** (0.009)	0.418*** (0.018)
Observations	478,540	34,945	418,570	140,625
[F-statistic]	[2,932.00]	[639.60]	[1,924.00]	[523.40]
Spouse share of couple dividends				
	22.773***	43.300***	13.539***	19.255***
S_{it}	(2.336)	(6.485)	(2.545)	(4.961)
First stage	0.491*** (0.011)	0.592*** (0.026)	0.447*** (0.012)	0.470*** (0.019)
Observations	295,030	31,500	245,070	99,625
[F-statistic]	[2,072.00]	[510.90]	[1,294.00]	[596.70]
Log family taxable income				
	0.016	-0.004	0.025	0.033
S_{it}	(0.026)	(0.044)	(0.034)	(0.088)
First stage	0.450*** (0.008)	0.605*** (0.024)	0.406*** (0.009)	0.418*** (0.018)
Observations	478,070	34,940	418,105	140,480
[F-statistic]	[2,932.00]	[639.50]	[1,924.00]	[523.70]
Log family tax liability				
	-0.076	-0.118	-0.018	-0.056
S_{it}	(0.068)	(0.086)	(0.091)	(0.201)
First stage	0.462*** (0.009)	0.604*** (0.024)	0.419*** (0.010)	0.430*** (0.019)
Observations	455,970	34,830	396,705	136,450
[F-statistic]	[2,950.00]	[636.50]	[1,930.00]	[535.60]

Notes: Both stages follow Equations (5) and (6). The shifting incentive S_{it} is defined in Equation (3) and is the first-stage dependent variable. Each second-stage outcome has its own sample and first stage. Family-clustered standard errors are in parentheses; first-stage F-statistics are in square brackets below observations. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$. The professional and non-professional columns use their respective classified subsamples. The multiple-business column includes spouses holding shares in more than one business. Samples differ across outcomes and columns.

Table C8: Tax-Incentive IV Estimates Under Alternative Sample and Treatment Definitions

	Main	Unbalanced	A3	A3B	E3
Log spouse dividends					
	1.299***	1.385***	2.278***	1.212***	1.158***
S_{it}	(0.193)	(0.142)	(0.271)	(0.227)	(0.190)
First stage	0.396*** (0.013)	0.416*** (0.009)	0.274*** (0.010)	0.452*** (0.017)	0.400*** (0.013)
Observations	239,735	414,860	248,455	112,625	243,055
[F-statistic]	[975.70]	[1,929.00]	[722.80]	[706.80]	[1,007.00]
Log owner dividends					
	-0.342***	-0.260***	-0.786***	-0.460***	-0.276**
S_{it}	(0.118)	(0.086)	(0.157)	(0.134)	(0.115)
First stage	0.508*** (0.012)	0.531*** (0.008)	0.356*** (0.010)	0.553*** (0.014)	0.518*** (0.012)
Observations	255,785	451,140	265,455	127,220	259,600
[F-statistic]	[1,927.00]	[3,949.00]	[1,341.00]	[1,569.00]	[1,991.00]
Spouse dividend receipt					
	0.243***	0.296***	0.396***	0.214***	0.216***
S_{it}	(0.026)	(0.019)	(0.036)	(0.029)	(0.026)
First stage	0.450*** (0.008)	0.469*** (0.006)	0.316*** (0.007)	0.520*** (0.010)	0.456*** (0.008)
Observations	478,540	904,010	492,450	225,970	483,540
[F-statistic]	[2,932.00]	[5,822.00]	[1,999.00]	[2,494.00]	[3,010.00]
Spouse share of couple dividends					
	22.773***	27.336***	45.586***	21.292***	18.867***
S_{it}	(2.336)	(1.704)	(3.512)	(2.734)	(2.257)
First stage	0.491*** (0.011)	0.510*** (0.008)	0.334*** (0.009)	0.533*** (0.013)	0.501*** (0.011)
Observations	295,030	528,375	306,355	148,015	299,395
[F-statistic]	[2,072.00]	[4,347.00]	[1,374.00]	[1,596.00]	[2,149.00]
Log family taxable income					
	0.016	0.003	-0.017	0.012	0.015
S_{it}	(0.026)	(0.019)	(0.033)	(0.027)	(0.026)
First stage	0.450*** (0.008)	0.469*** (0.006)	0.316*** (0.007)	0.520*** (0.010)	0.456*** (0.008)
Observations	478,070	902,500	491,975	225,745	483,065
[F-statistic]	[2,932.00]	[5,815.00]	[1,998.00]	[2,493.00]	[3,010.00]
Log family tax liability					
	-0.076	-0.105**	-0.190**	-0.053	-0.065
S_{it}	(0.068)	(0.047)	(0.085)	(0.068)	(0.067)
First stage	0.462*** (0.009)	0.482*** (0.006)	0.320*** (0.007)	0.525*** (0.011)	0.469*** (0.009)
Observations	455,970	853,425	469,695	217,680	460,915
[F-statistic]	[2,950.00]	[5,963.00]	[1,967.00]	[2,466.00]	[3,030.00]

Notes: Both stages follow Equations (5) and (6). The shifting incentive S_{it} is defined in Equation (3) and is the first-stage dependent variable. Each second-stage outcome has its own sample and first stage. Family-clustered standard errors are in parentheses; first-stage F-statistics are in square brackets below observations. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$. A3 and A3B are the ownership-based alternatives described in Appendix A.5. E3 assigns treated status using pre-2018 ownership and employment information while retaining the main control definition. The unbalanced column relaxes the balanced-panel restriction. These comparisons preserve qualitative dividend responses but do not impose identical coefficient magnitudes.

D Business Type and Ownership Comparisons

The following tables report the difference-in-differences estimates discussed in Section 6. The corresponding tax-incentive IV estimates appear in Table C7.

D.1 Professional and Non-Professional Corporations

Table D1: TC Income Responses by Corporation Type

	Non-professional		Professional	
	Log income	$1,000	Log income	$1,000
Spouse dividends	-0.283***	-4.996***	-1.210***	-24.875***
	(0.075)	(0.609)	(0.130)	(2.202)
Observations	199,880	418,760	25,465	34,945
Owner dividends	-0.004	1.861*	0.448***	20.384***
	(0.047)	(0.951)	(0.074)	(2.914)
Observations	212,680	418,760	27,275	34,945
Spouse total T4	-0.024	-2.615*	-0.125***	-7.047***
	(0.028)	(1.500)	(0.037)	(2.191)
Observations	329,330	418,760	28,355	34,945
Owner total T4	0.092***	1.972	0.041	5.220
	(0.032)	(1.771)	(0.039)	(3.202)
Observations	326,510	418,760	28,550	34,945
Spouse T4 from family business	0.027	-2.970**	-0.243***	-20.343***
	(0.077)	(1.204)	(0.065)	(2.946)
Observations	189,460	189,640	16,065	16,065
Owner T4 from family business	-0.144	-3.030	-0.032	13.978*
	(0.094)	(1.986)	(0.055)	(7.990)
Observations	156,335	156,510	13,050	13,060
Family taxable income	-0.025**	-4.772**	-0.004	-2.170
	(0.010)	(2.173)	(0.012)	(4.164)
Observations	418,295	418,760	34,940	34,945
Family tax liability	-0.005	-1.011	0.024	-0.185
	(0.027)	(0.932)	(0.024)	(1.704)
Observations	396,825	418,760	34,830	34,945
Dividend receipt				
Spouse receives dividends	-0.116***		-0.217***	
	(0.011)		(0.019)	
Observations	418,760		34,945	
Owner receives dividends	0.002		0.090***	
	(0.010)		(0.014)	
Observations	418,760		34,945	

Notes: TC coefficients compare treated and control families during 2018–2020 within each corporation type. Dollar outcomes are in thousands; receipt outcomes are zero-one indicators. Each outcome has its own sample. Family-clustered standard errors are in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

D.2 Multiple-Business Ownership

Table D2: Treatment–Control Income Responses by Spouse Business Ownership

	Main sample		Multiple businesses	
	Log income	$1,000	Log income	$1,000
Spouse dividend income	-0.753***	-12.391***	-0.383***	-6.274**
	(0.066)	(0.815)	(0.114)	(2.929)
Observations	239,795	478,735	87,780	140,655
Primary-owner dividend income	0.134***	7.124***	-0.120	4.397
	(0.037)	(1.149)	(0.095)	(3.214)
Observations	255,870	478,735	93,425	140,655
Spouse T4 income	-0.079***	-4.419***	-0.012	-2.163
	(0.020)	(1.136)	(0.047)	(4.858)
Observations	377,555	478,735	113,160	140,655
Primary-owner T4 income	0.153***	10.867***	0.086*	5.162
	(0.024)	(1.505)	(0.050)	(4.705)
Observations	374,695	478,735	117,250	140,655
Spouse T4 from family business	-0.133***	-11.503***	-0.421**	-20.120**
	(0.047)	(1.296)	(0.169)	(7.952)
Observations	215,465	215,660	36,070	36,090
Primary-owner T4 from family business	-0.040	9.729*	-0.217	2.373
	(0.051)	(5.174)	(0.151)	(9.356)
Observations	177,420	177,615	31,015	31,035
Family taxable income	-0.012*	0.928	-0.009	6.831
	(0.007)	(1.875)	(0.019)	(10.374)
Observations	478,260	478,735	140,510	140,655
Family tax liability	0.032*	0.225	0.055	5.059
	(0.017)	(0.755)	(0.039)	(4.471)
Observations	456,090	478,735	136,480	140,655
Dividend receipt				
Spouse receives dividends	-0.168***		-0.076***	
	(0.009)		(0.020)	
Observations	478,735		140,655	
Primary owner receives dividends	0.029***		0.030**	
	(0.007)		(0.014)	
Observations	478,735		140,655	

Notes: TC implementation-period estimates for 2018–2020. The main sample restricts the spouse to shares in one business; the comparison sample includes spouses with shares in multiple businesses. Dollar outcomes are in thousands and receipt is a zero-one indicator. Each outcome has its own sample. Family-clustered standard errors are in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.